

Intracranial region

The neurocranium (calvaria and cranial base) contains the brain, the intracranial portions of the cranial nerves, blood vessels, meninges and cerebrospinal fluid. Its walls are formed by parts of the frontal, ethmoid, parietal, sphenoid, temporal and occipital bones.

INTERNAL SURFACE OF CALVARIA

The internal surface of the calvaria is deeply concave. It includes most of the frontal and parietal bones and the squamous part of the occipital bone, variously united at the coronal, sagittal and lambdoid sutures (Fig. 28.1). With increasing age, these sutures become obliterated by a gradual process that begins on their intracranial surfaces. Inconstant foramina may occur in the parietal bones near the sagittal sulcus and anterior to the lambdoid suture; they admit emissary veins associated with the superior sagittal sinus. The internal surfaces of the frontal and parietal bones are grooved by furrows that house the frontal and

parietal branches of the middle meningeal vessels; the grooves contain the openings of minute channels that admit perforating vessels to the haemopoietic marrow within the diploic bone. Impressions for cerebral gyri are less distinct on the bones of the calvaria than they are on the cranial base.

The falx cerebri, a dural partition that separates the two cerebral hemispheres of the brain, is attached anteriorly to a backward-projecting anteromedian frontal crest. The crest exhibits a groove that widens as it passes back below the sagittal suture and becomes continuous with the sagittal sulcus, which houses the superior sagittal sinus. Irregular depressions, granular foveolae, which become larger and more numerous with age, lie on either side of the sulcus and usually house arachnoid granulations.

CRANIAL FOSSAE (ANTERIOR, MIDDLE, POSTERIOR)

The base of the cranial cavity is divided into three distinct fossae: the anterior, middle and posterior cranial fossae (Fig. 28.2). The floor of the anterior cranial fossa is at the highest level and the floor of the posterior fossa is at the lowest.

Detailed descriptions of the microsurgical anatomy of each region of the cranial base are beyond the scope of this book, and the interested reader is directed in the first instance to [Rhoton \(2007\)](#).

ANTERIOR CRANIAL FOSSA

The anterior cranial fossa is formed by the frontal, ethmoid and sphenoid bones, and supports the frontal lobes of the cerebral hemispheres. Its floor is composed of the orbital plate of the frontal bone, the cribriform plate and crista galli of the ethmoid bone, and the lesser wings, jugum sphenoidale, and prechiasmatic sulcus of the sphenoid.

A perforated plate of bone, the cribriform plate of the ethmoid bone, spreads across the midline between the orbital plates of the frontal bone and is depressed below them, forming part of the roof of the nasal cavity. Olfactory nerves pass from the nasal mucosa to the olfactory bulb of the brain through numerous small foramina in the cribriform plate. Anteriorly, a spur of bone, the crista galli, projects upwards between the cerebral hemispheres and serves as an attachment for the falx cerebri. A depression between the crista galli and the crest of the frontal bone is crossed by the frontoethmoidal suture. It bears the foramen caecum, which is usually a small blind-ended depression formed by the developmental involution of a dural diverticulum that extends through the foramen caecum to the columella; in adults, it occasionally accommodates a vein draining from the nasal mucosa to the superior sagittal sinus. Failure of the foramen caecum to close may cause nasal dermoid cysts and nasoethmoidal encephalocoeles in children.

The anterior ethmoidal nerve and vessels enter the cranial cavity where the cribriform plate meets the orbital part of the frontal bone. They then pass into the roof of the nasal cavity via a small foramen by the side of the crista galli. The posterior ethmoidal canal, which transmits the posterior ethmoidal nerve and vessels, opens at the junction of the posterolateral corner of the cribriform plate and jugum sphenoidale. The convex orbital plate of the frontal bone separates the brain from the orbit and bears impressions of cerebral gyri and small grooves for meningeal vessels. Posteriorly, it articulates with the anterior border of the lesser wing of the sphenoid laterally and the jugum sphenoidale medially. The jugum sphenoidale is a flat surface on the upper aspect of the sphenoid bone that connects the two lesser wings, also known as the planum sphenoidale. The limbus of the sphenoid bone is a prominent ridge on the body of the sphenoid that forms the posterior border of the jugum sphenoidale and the anterior border of the prechiasmatic sulcus. The latter is an osseous groove that runs transversely

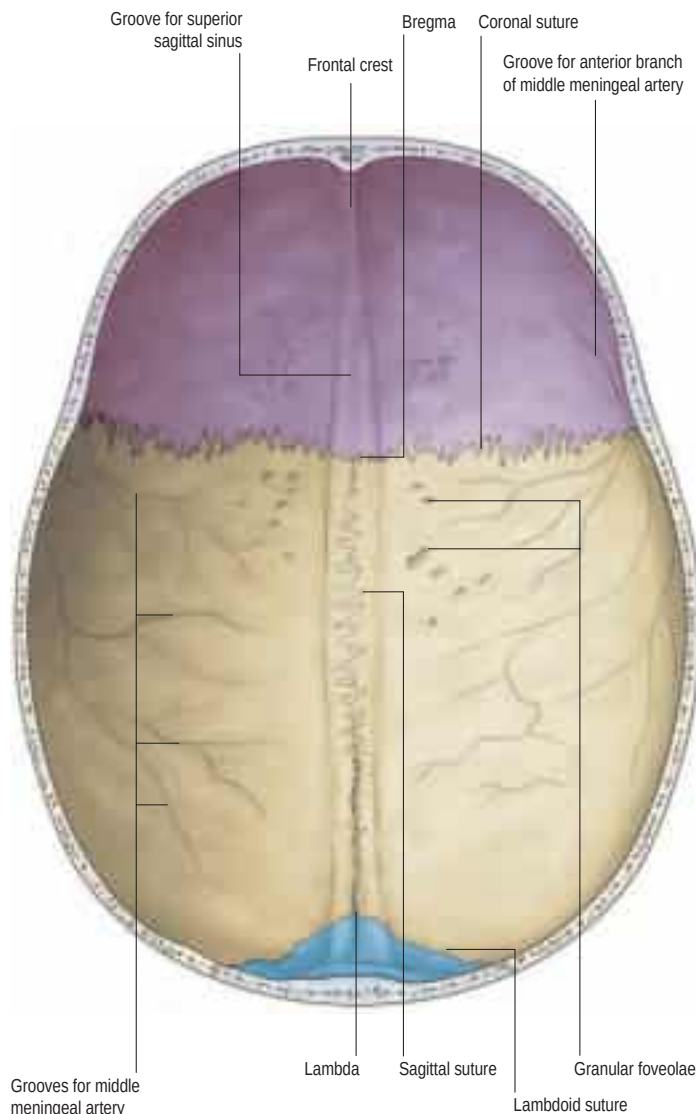


Fig. 28.1 The internal surface of the cranial vault. (With permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

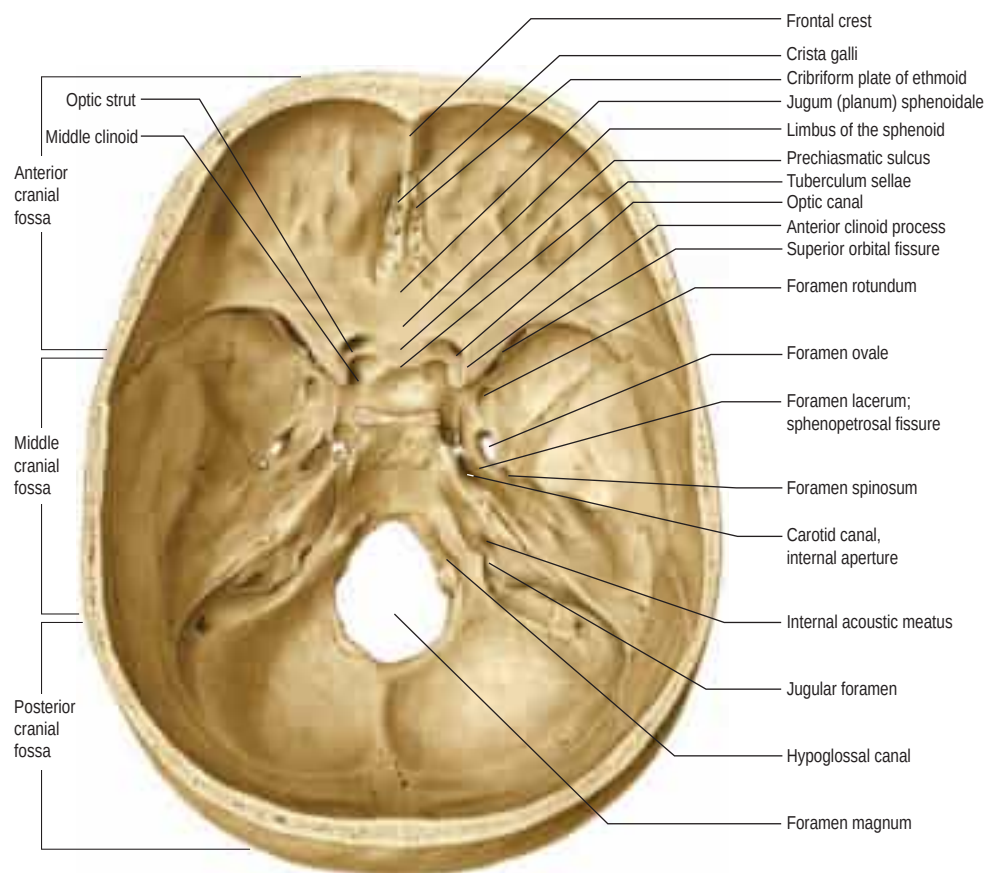


Fig. 28.2 The floor of cranial cavity showing the cranial fossae. (Adapted with permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer, 2013.)

between the optic canals. The optic chiasma usually lies above the prechiasmatic sulcus, while the medial orbital surface of the frontal lobes and olfactory tracts lie above the jugum sphenoidale (see Figs 28.2; 28.9).

The posterior boundary of the lesser wing forms the posterior boundary of the anterior cranial fossa; the medial end of the lesser wing constitutes the anterior clinoid process. The lesser wing and the anterior clinoid process join the sphenoid body by two roots that are separated by the optic canal. The anterior root, broad and flat, forms the roof of the optic canal and is continuous with the jugum sphenoidale, while the smaller and thicker posterior root, also known as the optic strut, joins the body of the sphenoid bone at the lateral aspect of the posterior bank of the prechiasmatic sulcus or tuberculum sellae (see Fig. 28.2). The optic strut of the anterior clinoid process separates the optic canal from the superior orbital fissure (Kerr et al 2012). Each optic canal transmits the optic nerve and ophthalmic artery.

MIDDLE CRANIAL FOSSA

The middle cranial fossa is formed by the sphenoid and temporal bones and supports the temporal lobes of the cerebral hemispheres. It is bounded in front by the posterior aspect of the lesser and greater wings of the sphenoid, behind by the superior border of the petrous part of the temporal bone, laterally by the squamous part of the temporal bone and greater wing of the sphenoid, and medially by the lateral aspect of the sphenoid body, including the carotid sulcus, sella turcica and dorsum sellae.

The middle cranial fossa communicates with the orbits by the superior orbital fissures, each bounded above by a lesser wing, below by a greater wing, and medially by the body of the sphenoid bone and optic strut. The fissures contain the oculomotor, trochlear and abducens nerves, and the lacrimal, frontal and nasociliary branches of the ophthalmic division of the trigeminal nerve, together with filaments from the internal carotid plexus (sympathetic), the superior and inferior ophthalmic veins, the orbital branch of the middle meningeal artery, and the recurrent branch of the lacrimal artery.

The central part of the cranial base, located between both middle cranial fossae, is formed by the body of the sphenoid bone, which contains the sphenoidal sinuses. The body of the sphenoid bone is deeply concave and houses the pituitary gland (hypophysis); it is therefore termed the pituitary (hypophysial) fossa, also known as the sella turcica because it resembles the shape of a Turkish saddle (Paluzzi et al 2012) (Videos 28.1–28.3). The face and floor of the sella turcica are

located within the sphenoidal air sinuses, which is why the most direct surgical route to the pituitary fossa is the trans-sphenoidal approach. The anterior edge of the sella is formed by a bony prominence known as the tuberculum sellae, which continues anteriorly with the prechiasmatic sulcus; the posterior boundary presents a vertical pillar of bone, the dorsum sellae. The superolateral angles of the dorsum are expanded as the posterior clinoid processes. A fold of dura, the diaphragma sellae, is attached to the anterior and posterior clinoid processes and roofs over the pituitary fossa. The cavernous sinus lies lateral to the pituitary fossa. The lateral wall of the body of the sphenoid contains a shallow carotid sulcus related to the internal carotid artery as it ascends from the carotid canal and runs through the cavernous sinus.

Each anterior clinoid process covers the roof of the cavernous sinus and the paraclinoid segment of the internal carotid artery as it leaves the cavernous sinus and pierces the dura to enter the intradural space. The anterior clinoid process gives attachment to the anterior petroclinoid ligament, which forms the free margin of the tentorium cerebelli, and the interclinoid ligament, which lies between the anterior and posterior clinoids. The middle clinoid process arises from the body of the sphenoid bone at the anterolateral margin of the sella turcica and may be connected to the anterior clinoid process by a thin osseous bar, formed by ossification of the caroticoclinoid ligament. The resulting caroticoclinoid foramen or ring surrounds the paraclinoid segment of the internal carotid artery (Fernandez-Miranda et al 2012). While anterior and posterior clinoids are always present, middle clinoids are not constant. A recent study revealed that the middle clinoid process was usually identifiable in fine-cut head computed tomogram–angiogram scans, and sometimes had at least one caroticoclinoid ring.

The greater wing of the sphenoid bone contains three consistent foramina and other small variable foramina. The foramen rotundum, situated just below the medial end of the superior orbital fissure, leads forwards into the medial aspect of the inferior orbital fissure and opens into the pterygopalatine fossa; it contains the maxillary nerve (second division of trigeminal nerve).

Behind the foramen rotundum is the foramen ovale, which transmits the mandibular nerve (third division of trigeminal nerve). The foramen spinosum lies just posterolateral to the foramen ovale and transmits the middle meningeal artery and veins; these vessels groove the floor and lateral wall of the middle cranial fossa. The foramen ovale and foramen spinosum open into the underlying infratemporal fossa. The small foramen of Vesalius sometimes occurs in skulls and lies antero-medial to the foramen ovale and lateral to the foramen rotundum; it transmits an emissary vein through which the cavernous sinus and pterygoid plexus communicate.

The foramen lacerum is a misnomer because it is a gap, not a true foramen, formed by the incomplete confluence of the petrosphenoidal and petroclival fissures. It may therefore also be called the sphenopetroclival synchondrosis. It lies posteromedial to the foramen ovale at the transition between the carotid sulcus of the sphenoid bone and the petrous carotid canal, bounded laterally by the lingula of the sphenoid and the petrolingual ligament extending to the petrous apex. The lower portion of the foramen lacerum is filled with fibrocartilage extending from the pharyngotympanic tube and basopharyngeal fascia. The upper portion of the foramen contains the laceral segment of the internal carotid artery and should be considered as the rostral extension of the petrous carotid canal (Tauber et al 1999). The only structures that enter or exit the intracranial cavity through the foramen lacerum are the pterygoid (Vidian) nerve and vessels, terminal branches of the ascending pharyngeal artery and small emissary veins connecting the extracranial pterygoid plexus with the intracranial cavernous sinus.

The superior border of the petrous part of the temporal bone separates the middle and the posterior cranial fossae, and is grooved by the superior petrosal sinus. The trigeminal impression, which accommodates the trigeminal ganglion, is situated on the anterior surface of the petrous part of the temporal bone near its apex. A smooth trigeminal notch leads into the impression and lies on the upper border of the petrous part, anteromedial to the groove for the superior petrosal sinus. At the level of the notch, the trigeminal nerve passes from the posterior to the middle cranial fossa, and separates the superior petrosal sinus from the underlying bone. The petrosphenoidal ligament (of Gruber) lies at the anterior end of the trigeminal notch, attached to a minute bony spicule, the petrous tubercle (Iaconetta et al 2003). The ligament runs in a superior, anterior and medial direction to attach to the posterior and lateral aspect of the posterior clinoid. The abducens nerve bends sharply across the upper petrous border and medially over the lateral edge of the dorsum sellae, and typically passes under the petrosphenoidal ligament to enter the cavernous sinus (Ozveren et al 2002). This interdural space (between dural layers), limited superiorly by the petrosphenoidal ligament, laterally by the petrous apex, and medially by the dorsum sellae, is known as Dorello's canal (Ambekar et al 2012). It is also defined as the petroclival venous confluence because it is filled with blood from the adjacent venous sinuses, namely: the inferior and superior petrosal, cavernous and basilar plexus (Barges-Coll et al 2010).

Lateral to the trigeminal impression, a narrow groove passes posterolaterally into the hiatus for the greater petrosal nerve (also known as the facial hiatus); the hiatus and groove for the lesser petrosal nerve lies even further laterally (Kakizawa et al 2007). Posterolateral to the facial hiatus, a rounded arcuate eminence is produced by the underlying superior semicircular canal. A line bisecting the angle subtended by the arcuate eminence and the greater petrosal nerve estimates the location of the roof of the internal acoustic meatus (internal auditory canal) in the middle fossa. The petrous segment of the internal carotid artery runs from the carotid foramen to the foramen lacerum within the carotid canal of the petrous part of the temporal bone. The greater petrosal nerve travels above and parallel to the petrous carotid canal, which may be dehiscent. The osseous portion of the pharyngotympanic tube and tensor tympani run parallel and just lateral to the petrous carotid canal within the floor of the middle fossa. The cochlea is situated within the petrous part of the temporal bone, just medial to the posterior aspect of the petrous carotid canal and anterior to the internal acoustic meatus and arcuate eminence. The portion of petrous bone that is limited anteriorly by the mandibular nerve, laterally by the greater petrosal nerve, medially by the superior petrosal sinus, and posteriorly by the internal acoustic meatus and cochlea is named the posteromedial or Kawase's triangle of the middle fossa. The surgical procedure known as anterior petrosectomy consists of removal of this bone in order to access the posterior fossa from the middle fossa approach (Kawase et al 1985) (Ch. 38).

The lateral surface of the petrous part of the temporal bone is formed by the tegmen tympani, a thin osseous lamina that separates the tympanic cavity (middle ear) from the middle fossa. Chronically elevated intracranial pressure may cause erosion of the tegmen tympani and spontaneous cerebrospinal fluid otorrhoea. The posterior part of the tegmen tympani forms the roof of the mastoid antrum, lateral to the arcuate eminence, and is also called the tegmen mastoideum. In young skulls, a petrosquamous suture may be visible at the lateral limit of the tegmen tympani but it is obliterated in adults. The tegmen tympani then turns down as the lateral wall of the osseous portion of the pharyngotympanic tube, and its lower border may appear in the squamotympanic fissure. Lateral to the anterior part of the tegmen tympani, the squamous part of the temporal bone is thinned over a small area that coincides with the deepest part of the mandibular fossa.

POSTERIOR CRANIAL FOSSA

The posterior cranial fossa is formed by the sphenoid, temporal and occipital bones, and contains the cerebellum, pons and medulla oblongata. It is bounded in front by the dorsum sellae, posterior aspects of the body of the sphenoid bone, and the basilar (or clival) part of the occipital bone; behind by the squamous part of the occipital bone; and laterally by the petrous and mastoid parts of the temporal bones and by the lateral (condylar) parts of the occipital bone.

The most prominent feature in the floor of the posterior cranial fossa is the foramen magnum in the occipital bone. The medulla oblongata exits the cranial cavity through the foramen magnum to continue as the spinal cord (Ch. 20). The cerebellum is fully contained within the posterior fossa. When the lower part of the cerebellum, the tonsils, descend abnormally through the foramen magnum, they may cause compression of the medulla at the level of the foramen magnum, a condition called tonsillar herniation or Chiari malformation.

The clivus is a sloping surface that extends from the sella turcica to the foramen magnum, and is formed successively by the dorsum sellae, the posterior part of the body of the sphenoid and the basilar (clival) part of the occipital bone. The sphenopetroclival synchondrosis is evident on the clivus of a growing child but not in adults. The clivus is located at the centre of the posterior skull base and separates the nasopharynx from the posterior cranial fossa. Cranial chordomas are malignant tumours that histologically are reminiscent of embryonic notochordal tissue and typically arise within the clival bone (Fernandez-Miranda et al 2014). The clivus reaches its final length at the age of 11 years in both sexes (Krmptić-Nemanić et al 2005). On each side, the clivus is separated from the petrous part of the temporal bone by a petro-occipital (or petroclival) fissure, filled by a thin plate of cartilage and extending from the foramen lacerum (extracranially) and cavernous sinus (intracranially) to the jugular foramen. The intracranial margins of the fissure are grooved by the inferior petrosal sinus. Chondrosarcomas are malignant tumours composed of transformed cells that produce cartilage; their most common intracranial location is precisely the petroclival fissure.

The jugular foramen is a large opening located between the temporal and occipital bones at the posterior end of the petro-occipital fissure, above and lateral to the foramen magnum. The lower border of the jugular foramen is smooth and is formed by the lateral part of the occipital bone. The upper border is sharp and irregular, and is formed by the petrous part of the temporal bone. It contains a notch, the intrajugular process, which divides the foramen into a large posterolateral compartment, the sigmoid part, which receives the drainage of the sigmoid sinus; and a small anteromedial compartment, the petrosal part, which receives the drainage of the inferior petrosal sinus. The accessory, vagus and glossopharyngeal nerves course in the medial side of the intrajugular process; they lie between the petrosal part (which contains the inferior petrosal sinus accompanied by a meningeal branch of the ascending pharyngeal artery) and the sigmoid part (which contains the sigmoid sinus accompanied by a meningeal branch of the occipital artery). The cochlear canaliculus or aqueduct, which contains the perilymphatic 'duct', is sited in the deepest part of the intrajugular process. The jugular foramina are typically asymmetric: the right foramen is usually larger than the left, reflecting the right dominance of the sigmoid sinus and jugular vein.

The jugular tubercle is a rounded prominence located at the junction of the basilar and lateral parts of the occipital bone, just medial to the jugular foramen. The accessory, vagus and glossopharyngeal nerves cross the posterior portion of the tubercle on their way to the jugular foramen. The hypoglossal (anterior condylar) canal lies just inferior to the jugular tubercle, superior to the occipital condyle, and medial to and below the lower border of the jugular foramen (Morera et al 2010). It transmits the hypoglossal nerve and its recurrent branch, together with the meningeal branch of the ascending pharyngeal artery and an emissary vein that links the (intracranial) basilar plexus with the (extracranial) internal jugular vein. The canal may be divided by bony septa for different rootlets of the hypoglossal nerve. The occipital condyles form the lateral (or condylar) part of the occipital bone, and they lie within the anterior and lateral aspect of the foramen magnum. On each side, their medial surface bears a tubercle that gives attachments to the alar ligaments of the odontoid process, and their inferior surface is convex and articulates with the upper surface of the lateral mass of the atlas. The posterior condylar canal, when present, is located in the extracranial surface behind and above each occipital condyle; its internal orifice is posterolateral to that of the hypoglossal canal and contains the supracondylar vein and a meningeal branch of the occipital artery.

The posterior surface of the petrous part of the temporal bone forms much of the anterolateral wall of the posterior cranial fossa. It contains the internal acoustic meatus, which lies anterosuperior to the jugular foramen, and transmits the facial and vestibulocochlear nerves, the nervus intermedius and labyrinthine vessels. A small subarcuate fossa lined with dura mater lies posteriorly between the internal acoustic meatus and the opening of the vestibular aqueduct, and houses the saccus and ductus endolymphaticus and a small artery and vein. The suprameatal tubercle is a bony prominence located superiorly between the internal acoustic meatus and the petrous ridge (Seoane and Rhoton 1999).

The mastoid part of the temporal bone lies behind the petrous part of the temporal bone in the lateral wall of the posterior cranial fossa. Anteriorly, it is grooved by a wide sigmoid sulcus that runs forwards and downwards, then downwards and medially, and finally forwards to the jugular foramen; it contains the sigmoid sinus. Superiorly, where the groove touches the mastoid angle of the parietal bone, it becomes continuous with a groove that transmits the transverse sinus, and then crosses the parietomastoid suture and descends behind the mastoid antrum. A mastoid foramen, which transmits an emissary vein from the sigmoid sinus and a meningeal branch of the occipital artery, may be sited here. The lowest part of the sigmoid sulcus crosses the occipitomastoid suture and grooves the jugular process of the occipital bone. The right sigmoid sulcus is usually larger than the left.

The squamous part of the occipital bone displays a median internal occipital crest. This runs posteriorly from the foramen magnum to an internal occipital protuberance, gives attachment to the falx cerebelli, and may be grooved by the occipital sinus. The internal occipital protuberance is close to the confluence of the sinuses. It is grooved bilaterally by the transverse sinuses, which curve laterally with an upward convexity to the mastoid angles of the parietal bones. The groove for the transverse sinus is usually deeper on the right, where it is generally a continuation of the superior sagittal sinus, while on the left it is frequently a continuation of the straight sinus. On both sides, the transverse sulcus is continuous with the sigmoid sulcus. Below the transverse sulcus, the internal occipital crest separates two shallow fossae that house the cerebellar hemispheres. The posterior clinoid process and the margins of the grooves for the transverse and superior petrosal sinuses all provide anchorage for the attached margin of the tentorium cerebelli.

MENINGES

Three concentric membranes, the meninges, envelop the brain and spinal cord. They provide support and protection for the delicate tissues they surround. The individual layers, in order from outside to inside, are the dura mater (pachymeninx), arachnoid mater and pia mater. The dura is an opaque, tough, fibrous coat that incompletely divides the cranial cavity into compartments and accommodates the dural venous sinuses. The arachnoid is much thinner than the dura and is mostly translucent. It loosely surrounds the brain, cranial nerves and vessels, spanning fissures and sulci. The pia mater is a transparent, microscopically thin membrane that follows the contours of the brain and is closely adherent to its surface.

The dura is separated from the arachnoid by a narrow subdural space. The arachnoid is separated from the pia by the subarachnoid space, which varies greatly in depth; the larger expanses are termed subarachnoid cisterns. The subarachnoid space contains cerebrospinal fluid (CSF), which is secreted by the choroid plexuses of the cerebroventricular system. CSF circulates within the subarachnoid space and is reabsorbed into the venous system through arachnoid villi and granulations associated with the dural venous sinuses.

Cranial and spinal meninges are continuous through the foramen magnum. The cranial meninges are described in this section and the spinal meninges are described on page 764.

DURA MATER

Dura mater is thick, dense and fibrous. It is predominantly acellular, and consists mainly of densely packed fascicles of collagen fibres arranged in laminae. The fascicles run in different directions in adjacent laminae, producing a lattice-like appearance that is particularly obvious in the tentorium cerebelli and around the defects or perforations that sometimes occur in the anterior portion of the falx cerebri.

The cranial dura, which lines the cranial cavity, differs from the spinal dura mainly in its relationship to the surrounding bones. It has an inner, or meningeal, layer and an outer, or endosteal, layer. As a

general rule, the meningeal layer covers neural tissues, while the endosteal layer adheres to the osseous structures. These layers are united except where they separate to enclose the venous sinuses that drain blood from the brain. There is little histological difference between the endosteal and meningeal layers. Both contain fibroblasts, and the endosteal layer also contains osteoblasts. Focal calcification may occur in the falx cerebri and near the superior sagittal sinus.

The endosteal layer of dura adheres to the internal surfaces of the cranial bones, particularly at the sutures and the cranial base, and around the foramen magnum. Fibrous bands pass from the dura into the bones, and it is difficult to remove the dura from the suture lines in young skulls. However, as the suture lines fuse, the dura becomes separated from them. The endosteal layer of the dura is continuous with the pericranium through the cranial sutures and foramina and with the orbital periosteum through the superior orbital fissure. The meningeal layer provides tubular sheaths for the cranial nerves as they pass out through the cranial foramina, and these sheaths fuse with the epineurium as the nerves emerge from the skull. The dural sheath of the optic nerve is continuous with the ocular sclera. The dura fuses with the adventitia of major vessels, such as the internal carotid and vertebral arteries, at sites where they pierce it to enter the cranial cavity.

The inner, meningeal, layer of dura is closely applied to the arachnoid over the surface of the brain. However, the two membranes are physically joined only at sites where veins pass from the brain into venous sinuses, e.g. the superior sagittal sinus.

Extradural haematoma

The anatomical organization of the dura, and its relationships to the major venous sinuses, sutures and blood vessels, are clinically significant. Separation of the dura from the cranial bones requires significant force, and consequently happens when high-pressure arterial bleeding takes place into the extradural or epidural space. This can result from damage to any meningeal vessel, commonly following skull fracture. The classic site for such injury is along the course of the middle meningeal artery, where a direct blow causing a fracture of the temporal and/or parietal bones can rupture the artery and cause rapid collection of blood into the extradural space. An extradural haematoma therefore acts as a rapidly expanding intracranial mass lesion that causes acute brain compression and displacement; it is a classic medical emergency that requires immediate diagnosis and surgery through a craniotomy for epidural blood clot evacuation and coagulation of the ruptured vessel (see Fig. 28.12).



Dural partitions

The meningeal layer of the dura is reflected inwards to form four septa, namely: the falx cerebri, falx cerebelli, diaphragma sellae and tentorium cerebelli, which partially divide the cranial cavity into compartments.

Falx cerebri

The falx cerebri is a strong, crescent-shaped sheet that lies in the sagittal plane and occupies the longitudinal fissure between the two cerebral hemispheres (Fig. 28.3). The crescent is narrow in front, where the falx is fixed to the crista galli, and broad behind, where it blends with the tentorium cerebelli; the straight sinus runs along this line of attachment to the tentorium (see Fig. 28.3). The anterior part of the falx is thin and may have a number of irregular perforations. Its convex upper margin is attached to the internal cranial surface on each side of the midline, as far back as the internal occipital protuberance. The superior sagittal sinus runs in a cranial groove within the dura along this margin; the falx is attached to the lips of this groove. The lower edge of the falx is free and concave, and contains the inferior sagittal sinus.

Falx cerebelli

The falx cerebelli is a small midline fold of dura mater that lies below the tentorium cerebelli and projects forwards into the posterior cerebellar notch between the cerebellar hemispheres. Its base is directed upwards and is attached to the posterior part of the inferior surface of the tentorium cerebelli in the midline. Its posterior margin contains the occipital sinus and is attached to the internal occipital crest. The lower apex of the falx cerebelli frequently divides into two small folds, which disappear at the sides of the foramen magnum.

Diaphragma sellae

The diaphragma sellae is a small, circular, horizontal sheet of dura mater. It forms a roof over the sella turcica and covers the pituitary gland. The infundibulum (also known as pituitary stalk) passes into the pituitary fossa through a central opening in the diaphragma. There is wide individual variation in the size of this opening. The diaphragma

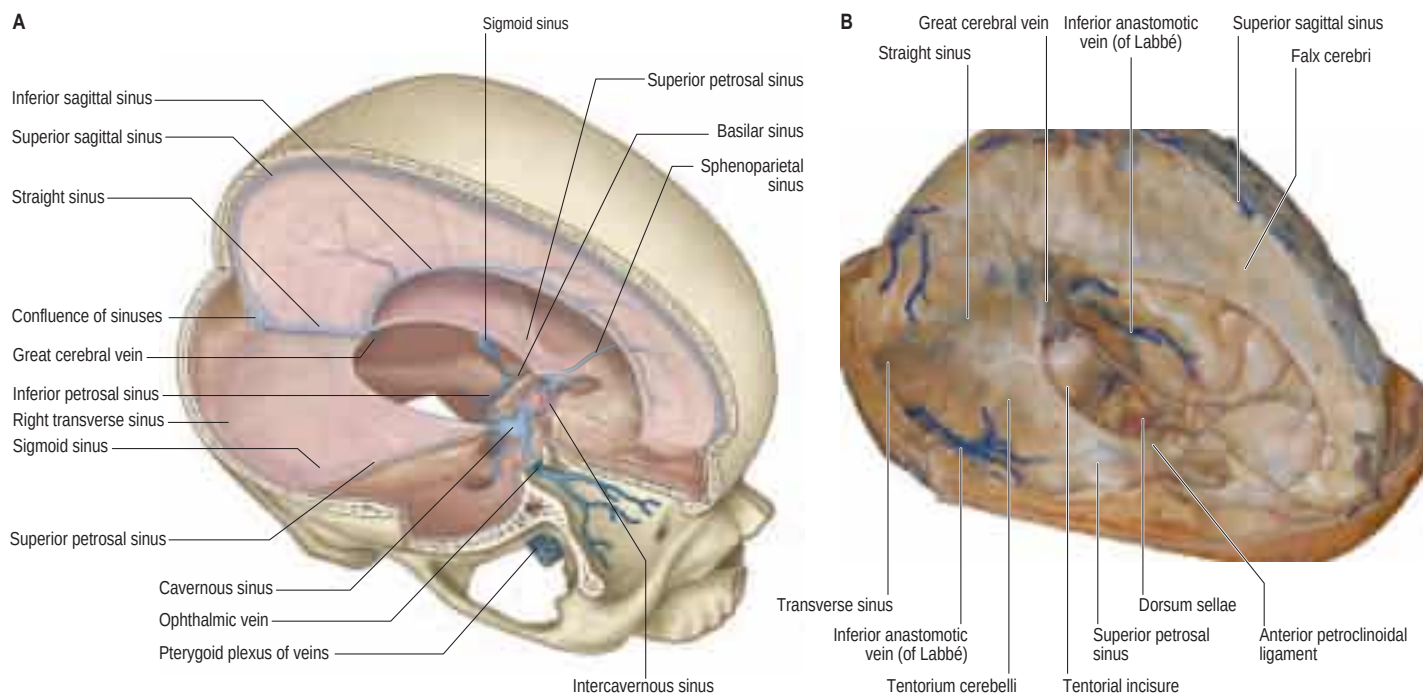


Fig. 28.3 **A**, The cerebral dura mater, its reflections and associated major venous sinuses. **B**, An anatomical dissection of a silicon-injected specimen. (A, Adapted from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

sellae is an important landmark structure in pituitary surgery because large pituitary tumours extend above it and may adopt a characteristic dumbbell shape. A trans-sphenoidal approach is currently the preferred option for accessing pituitary tumours, irrespective of whether there is suprasellar extension beyond the diaphragma sellae.

Tentorium cerebelli

The shape of the tentorium cerebelli (see Fig. 28.3B) is reminiscent of a single-poled tent, from which its name is derived. It lies between the cerebellum and the occipital lobes of the cerebral hemispheres, and divides the cranial cavity into supratentorial and infratentorial compartments, which contain the forebrain and hindbrain, respectively. Its concave anterior edge is free and separated from the dorsum sellae of the sphenoid bone by a large curved hiatus, the tentorial incisure or notch, which is filled by the midbrain and the anterior part of the superior aspect of the cerebellar vermis. The convex outer limit of the tentorium is attached posteriorly to the lips of the transverse sulci of the occipital bone and to the posterior inferior angles of the parietal bones, where it encloses the transverse sinuses. Laterally, it is attached to the superior borders of the petrous parts of the temporal bones, where it contains the superior petrosal sinuses. On each side, near the apex of the petrous temporal bone, the lower layer of the tentorium is evaginated anterolaterally under the superior petrosal sinus to form a recess, Meckel's cave, between the endosteal and meningeal layers in the middle cranial fossa. The cave is entered by the posterior root of the trigeminal nerve and contains cerebrospinal fluid and the trigeminal ganglion; the evaginated meningeal layer fuses in front with the anterior part of the ganglion.

The arrangement of the dura mater in the central part of the middle cranial fossa is complex. The tentorium forms the medial part of the floor of the middle cranial fossa. On both sides, the rim of the tentorial incisure is attached to the apex of the petrous temporal bone and continues forwards as a ridge of dura, known as the anterior petroclinoid ligament, to attach to the anterior clinoid process. This ligament marks the junction of the roof and the lateral wall of the cavernous sinus. The periphery of the tentorium cerebelli (attached to the superior border of the petrous temporal bone), crosses under the free border of the tentorial incisure at the apex of the petrous temporal bone, and continues forwards to the posterior clinoid process as a rounded ridge of the dura mater known as the posterior petroclinoid ligament. The dural extension between the anterior and posterior petroclinoid ligaments forms the roof of the cavernous sinus. On either side, it is pierced superiorly by the oculomotor nerve and behind by the trochlear nerve, which proceed anteroinferiorly into the lateral wall of the cavernous sinus.

In the anteromedial part of the middle cranial fossa, the dura ascends as the lateral wall of the cavernous sinus, reaches the ridge produced by the anterior petroclinoid ligament, and runs medially as the roof of

the cavernous sinus, where it is pierced by the internal carotid artery. The interclinoid ligament, between the anterior and posterior clinoid processes, forms the medial limit of the roof of the cavernous sinus and continues medially with the diaphragma sellae. At, or just below, the opening in the diaphragma for the pituitary stalk, the dura of the diaphragma and the suprasellar arachnoid blend with each other and with the capsule of the pituitary gland; the subarachnoid space does not extend into the sella turcica.

Transtentorial coning

Normally, the arrangement of dural partitions such as the falx cerebri and tentorium cerebelli may help to stabilize the brain within the cranial cavity. However, when there is focal brain swelling or a focal space-occupying lesion within the brain or cranial cavity, mass effect and raised intracranial pressure may cause the brain to herniate under the falx cerebri or, more significantly, through the tentorial incisure. In this case, the medial temporal lobe, and particularly the uncus, will compress the oculomotor nerve, midbrain and the posterior cerebral arteries. This life-threatening event and neurosurgical emergency, occurring in patients with supratentorial space-occupying lesions, is known as transtentorial uncus herniation. Similarly, space-occupying lesions in the smaller infratentorial compartment may cause upward herniation of the cerebellar vermis through the tentorial hiatus (upward transtentorial herniation) or downward herniation of the cerebellar tonsils through the foramen magnum (tonsillar herniation) (see Fig. 21.20); these neurosurgical emergencies require a suboccipital craniectomy.

Innervation of the cranial dura mater

The innervation of the cranial dura mater is derived mainly from the three divisions of the trigeminal nerve, the second and third cervical spinal nerves, and the cervical sympathetic trunk (Fig. 28.4). Less well-established meningeal branches have been described arising from the vagus and hypoglossal nerves, and possibly from the facial and glossopharyngeal nerves.

In the anterior cranial fossa, the dura is innervated by meningeal branches of the anterior and posterior ethmoidal nerves and anterior filaments of the meningeal rami of the maxillary (nervus meningeus medius) and mandibular (nervus spinosus) divisions of the trigeminal nerve. Nervi meningeus medius and spinosus are, however, largely distributed to the dura of the middle cranial fossa, which also receives filaments from the trigeminal ganglion. The nervus spinosus re-enters the cranium through the foramen spinosum with the middle meningeal artery, and divides into anterior and posterior branches that accompany the main divisions of the artery and supply the dura mater in the middle cranial fossa and, to a lesser extent, the anterior fossa and calvarium. The anterior branch communicates with the meningeal branch of the

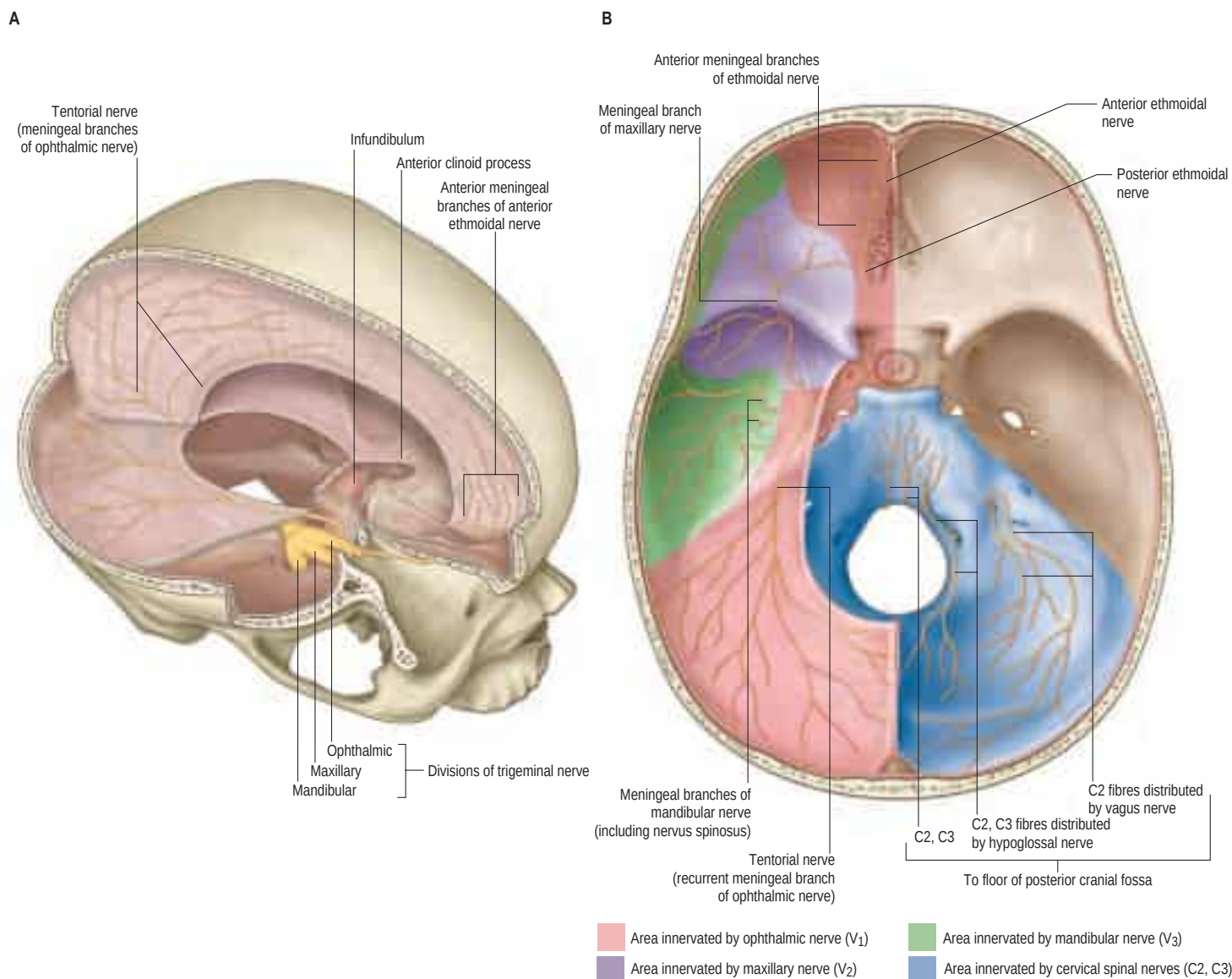


Fig. 28.4 The innervation of the cranial meninges.

maxillary nerve; the posterior branch also supplies the mucous lining of the mastoid air cells. The nervus spinosus contains sympathetic postganglionic fibres from the middle meningeal plexus. The nervus tentorii, a recurrent branch of the intracranial portion of the ophthalmic division of the trigeminal, supplies the supratentorial falx cerebri and the tentorium cerebelli. Intraoperative mechanical stimulation of the falx may trigger the trigeminocardiac reflex (Bauer et al 2005). The dura in the posterior cranial fossa is innervated by ascending meningeal branches of the upper cervical nerves, which enter through the anterior part of the foramen magnum (second and third cervical nerves) and through the hypoglossal canal and jugular foramen (first and second cervical nerves).

Meningeal branches from the vagus apparently start from the superior vagal ganglion and are distributed to the dura mater in the posterior cranial fossa. Those from the hypoglossal leave the nerve in its canal and supply the diploë of the occipital bone, the dural walls of the occipital and inferior petrosal sinuses, and much of the floor and anterior wall of the posterior cranial fossa. These meningeal rami may not contain vagal or hypoglossal fibres but ascending, mixed sensory and sympathetic fibres from the upper cervical nerves and superior cervical sympathetic ganglion. All meningeal nerves contain a postganglionic sympathetic component, either from the superior cervical sympathetic ganglion or by communication with its perivascular intracranial extensions. The role of the autonomic nerve supply of the cranial dura mater is uncertain.

Sensory nerve endings are restricted to the dura mater and cerebral blood vessels, and are not found in either the brain itself, or in the arachnoid or pia mater. Stimulation of these nerve endings causes pain, as evidenced during awake craniotomy procedures, and is the basis of certain forms of headache. The density of dural innervation, particularly around the dural venous sinuses, increases during the later part of ges-

tation, peaking at term and subsequently decreasing during the first year of postnatal life (Davidson et al 2012).

Dural venous sinuses

Dural venous sinuses form a complex network of venous channels that drain blood from the brain and cranial bones (see Figs 28.3, 19.11; Figs 28.5–28.6). They lie between the endosteal and meningeal layers of dura mater, are lined by endothelium and have no valves; their walls are devoid of muscular tissue (Kılıç and Akakin 2008). They develop initially as venous plexuses; most adult sinuses preserve a plexiform arrangement (to a variable degree), rather than being simple vessels with a single lumen. Plexiform arrays of small veins adjoin the superior and inferior sagittal and straight sinuses, and, less frequently, the transverse sinuses; ridges of 'spongy' venous tissue, known as venous lacunae, often project into the lumina of the superior sagittal and transverse sinuses (Browder and Kaplan 1976, Kaplan and Browder 1976).

The structure of cranial venous sinuses, their plexiform nature and their wide connections with cerebral and cerebellar veins vary considerably, particularly in earlier years, e.g. in infancy the falx cerebelli may contain large plexiform channels and venous lacunae that augment the occipital sinus. These variations must be established for the individual by catheter-based angiography, magnetic resonance (MR) venography or computed tomographic (CT) venography, when clinical necessity arises. The existence of arteriovenous shunts has been inferred from the demonstration of a connection between the middle meningeal arteries and the superior sagittal sinus (Browder and Kaplan 1976). Dural arteriovenous fistulae are thought to be acquired lesions that form in an area of thrombosis within a sinus. If the sinus remains completely thrombosed, venous drainage from these lesions takes place through

A

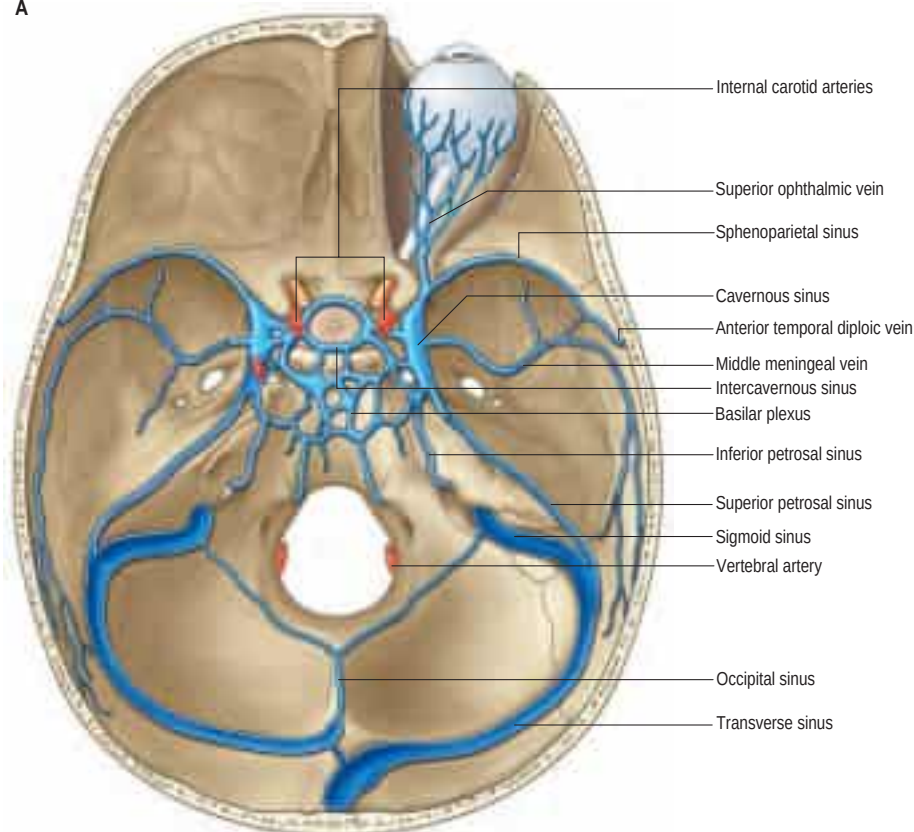


Fig. 28.5 The major venous sinuses at the base of the skull. **A**, The sinuses coloured dark blue have been opened up.

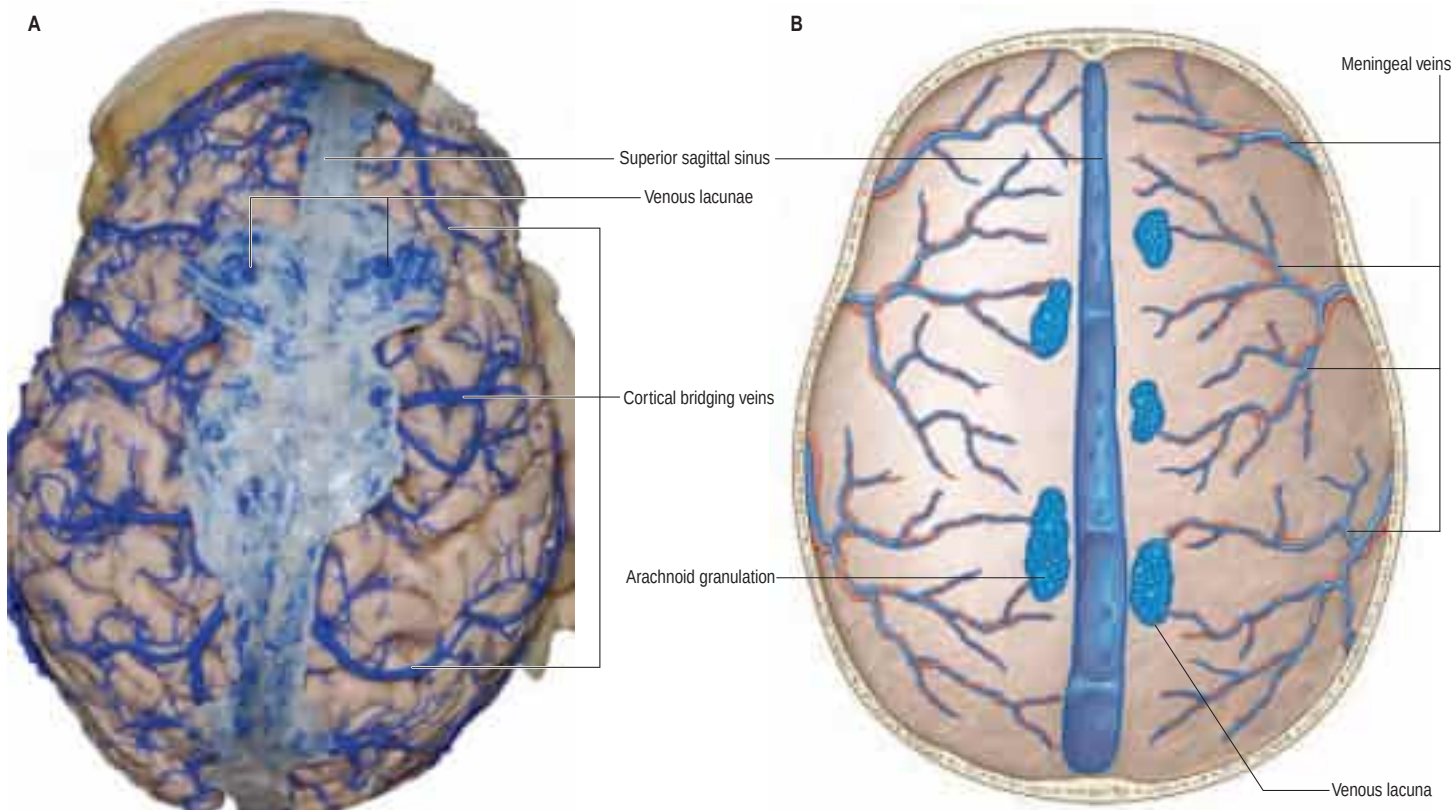


Fig. 28.6 The superior sagittal sinus and venous lacunae. **A**, An anatomical dissection of a silicon-injected specimen. **B**, The superior sagittal sinus opened up after removal of the cranial vault. Note the fibrous bands that cross the sinus from two of the venous lacunae. (A, From the Rhoton Collection.)

cortical veins, or, if the sinus is partially open, venous drainage is usually into the involved sinus. When the sinus remains closed, the elevated pressure within the cortical veins receiving the arterial flow carries the risk of haemorrhage.

The named sinuses are the superior and inferior sagittal, straight, transverse, sigmoid, occipital, cavernous, intercavernous, superior and inferior petrosal, sphenoparietal, basilar and marginal.

Superior sagittal sinus

The superior sagittal sinus runs in the attached convex margin of the falx cerebri. It grooves the internal surface of the frontal bone, the adjacent margins of the two parietal bones and the squamous part of the occipital bone (see Figs 28.3, 28.6, 19.11). It begins near the crista galli, a few millimetres posterior to the foramen caecum, and receives primary tributaries from cortical veins of the superior part of the frontal,

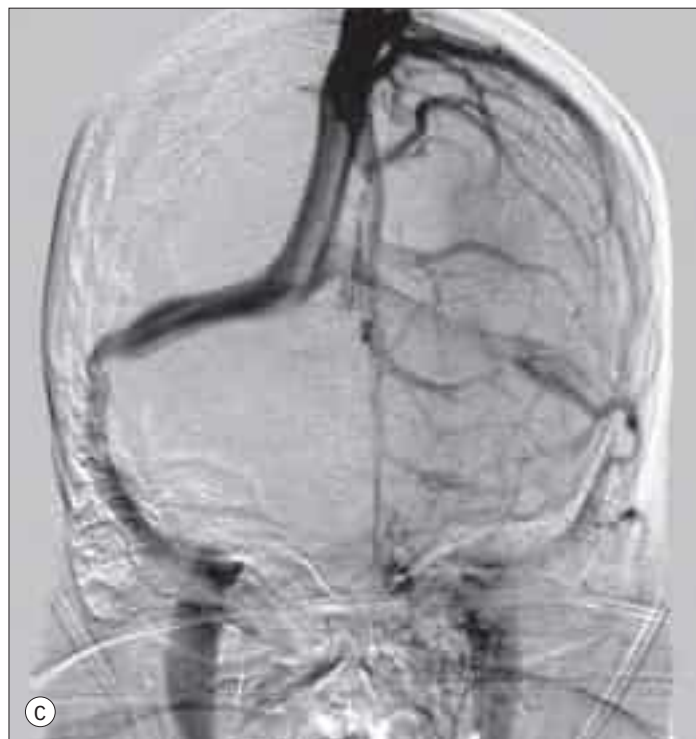
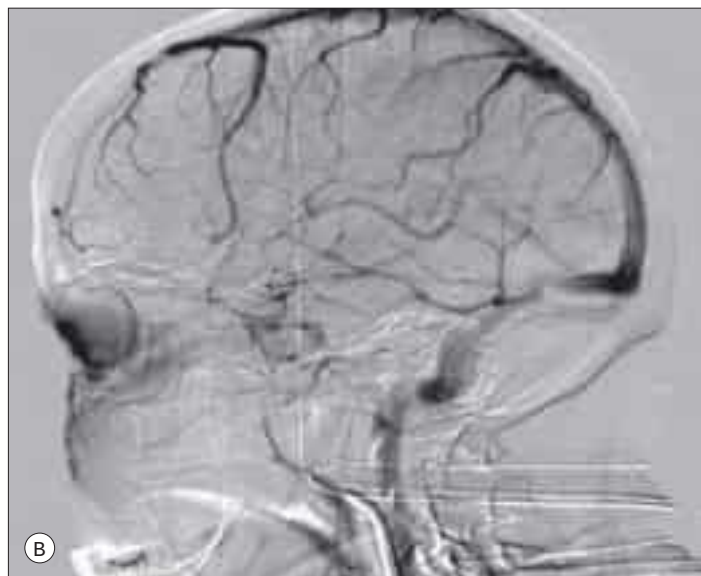


Fig. 28.5 B–C, Digitally subtracted venograms, venous phase. **B,** Sagittal view. **C,** Posterior view; note the asymmetry of the transverse sinuses.

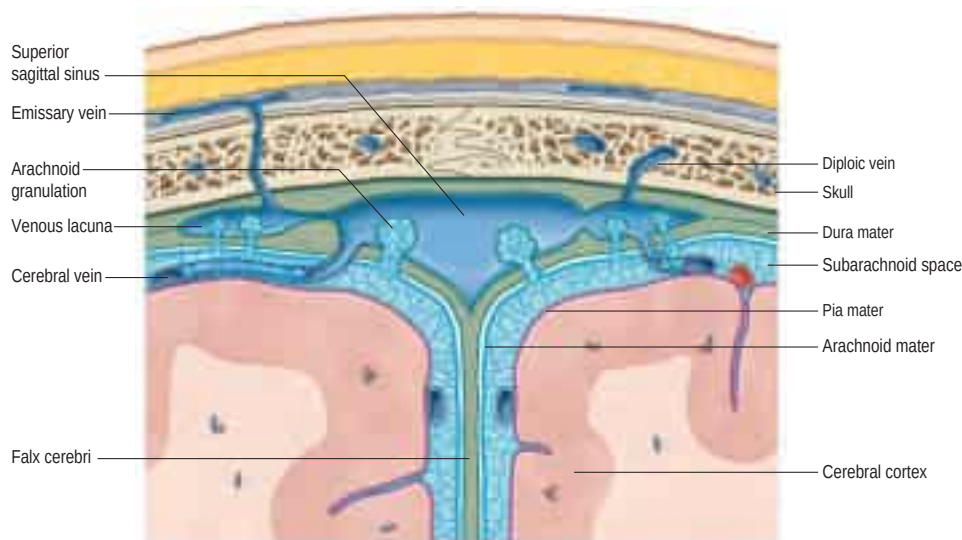


Fig. 28.7 A coronal section through the vertex of the skull to show the relationships between the superior sagittal sinus, meninges and arachnoid granulations. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

parietal and occipital lobes, and from the anterior part of the orbital surface of the frontal lobe. The sinus is triangular in cross-section, with the apex directed downwards and continuous with the falx cerebri. Its lumen is invaded in its intermediate third by projections from its dural walls, which may divide its lumen into superior and inferior channels. It is narrow anteriorly, and widens gradually to approximately 1 cm in diameter as it runs backwards. At its posterior end, the sinus enters the confluence of the sinuses (also known as the torcular herophili or the torcula, although the term actually describes the bony gutter in which the confluence lies), which is situated to one side (usually the right) of the internal occipital protuberance (see Fig. 28.3). At the confluence of the sinuses, the superior sagittal sinus usually deviates to become continuous with the right transverse sinus, but it also usually connects with the occipital and contralateral transverse sinuses. The size and degree of communication of the channels meeting at the confluence are highly variable; any sinus involved may be duplicated, narrowed or widened near the confluence.

There are usually two or three lateral venous lacunae on each side of the midline, named frontal (small), parietal (large) and occipital (intermediate) lacunae; they tend to become confluent in the elderly, producing a single elongated lacuna on each side. Fine fibrous bands cross the lacunae, and numerous arachnoid granulations project into them (see Fig. 28.6; Fig. 28.7). The lacunae mainly drain diploic and meningeal veins. The cortical veins typically pass beneath the lacunae on their way to the sinus. The largest cortical vein that connects the superficial Sylvian (middle cerebral) vein and the superior sagittal sinus is the superior anastomotic vein (vein of Trolard); it often runs in the precentral, central or postcentral sulci. Near its posterior end, the superior sagittal sinus receives veins from the pericranium that pass through parietal foramina. Acute and complete thrombosis of the superior sagittal sinus is an extremely severe condition causing acute elevation of the intracranial pressure and herniation. Slow and progressive occlusion of the sinus, as is typical for sagittal meningiomas, may be compensated by the development of collateral venous drainage with no clinical consequences.

Inferior sagittal sinus

The inferior sagittal sinus is located in the posterior half or two-thirds of the free margin of the falx cerebri (see Fig. 28.3). It increases in size posteriorly, ends in the straight sinus, and receives veins from the falx and sometimes from the medial surfaces of the cerebral hemispheres (anterior pericallosal veins).

Straight sinus

The straight sinus lies in the junction of the falx cerebri with the tentorium cerebelli (see Figs 28.3, 19.11). It runs posteroinferiorly as a continuation of the inferior sagittal sinus and drains into the transverse sinus, most commonly into the left one. It is not (or is only tenuously) continuous with the superior sagittal sinus. Its tributaries include the great cerebral vein (vein of Galen), which is formed by the convergence of the internal cerebral and basal veins, and some superior cerebellar veins. The latter may drain first into a short sinus within the tentorium, known as the medial tentorial sinus.

Transverse sinus

The transverse sinuses begin at the confluence of sinuses and attach to the internal occipital protuberance (see Figs 28.3, 28.5). One, usually

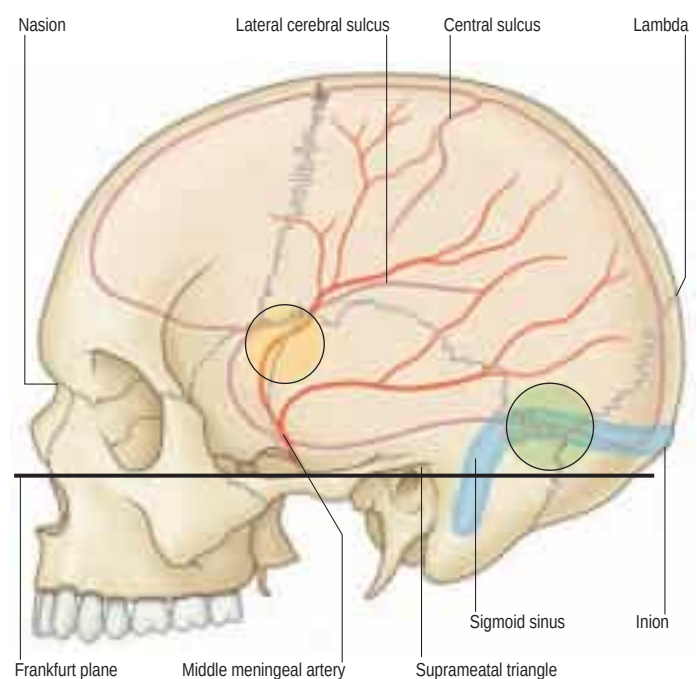


Fig. 28.8 The relations of the brain, the middle meningeal artery and the transverse and sigmoid sinuses to the surface of the skull. The area enclosed in the yellow circle (including the pterion) is used when performing a pterional craniotomy over the frontal branch of the middle meningeal artery and lateral Sylvian fissure; the area enclosed in the green circle is used when performing a retrosigmoid craniotomy extending to the confluence of the transverse and sigmoid sinuses.

the right, is directly continuous with the superior sagittal sinus, and the other with the straight sinus. The right transverse sinus is thus typically larger and drains blood from the superficial parts of the brain, while the left transverse sinus mainly drains blood from the deep parts of the brain. On either side, the sinuses run in the attached margin of the tentorium cerebelli, first on the squama of the occipital bone, then on the mastoid angle of the parietal bone. Each follows a gentle antero-lateral curve, increasing in size as it does so, to the posterolateral part of the petrous temporal bone. Here it turns down as a sigmoid sinus, which ultimately becomes continuous with the internal jugular vein. The transverse sinuses receive tributaries from the lateral temporal surface and basal surface of the temporal and occipital lobes. The inferior anastomotic vein (vein of Labbé) is the largest vein connecting the veins of the Sylvian fissure with the transverse sinus. All tributaries of each transverse sinus may drain first into a lateral tentorial sinus. The superior petrosal sinuses drain into the transition between the transverse and sigmoid sinuses on either side (see Figs 28.3, 28.5).

Sigmoid sinus

The sigmoid sinuses are continuations of the transverse sinuses, beginning where these leave the tentorium cerebelli (see Fig. 28.5; Fig. 28.8). Each sigmoid sinus curves inferomedially in a groove on the mastoid process of the temporal bone, crosses the jugular process of the occipital

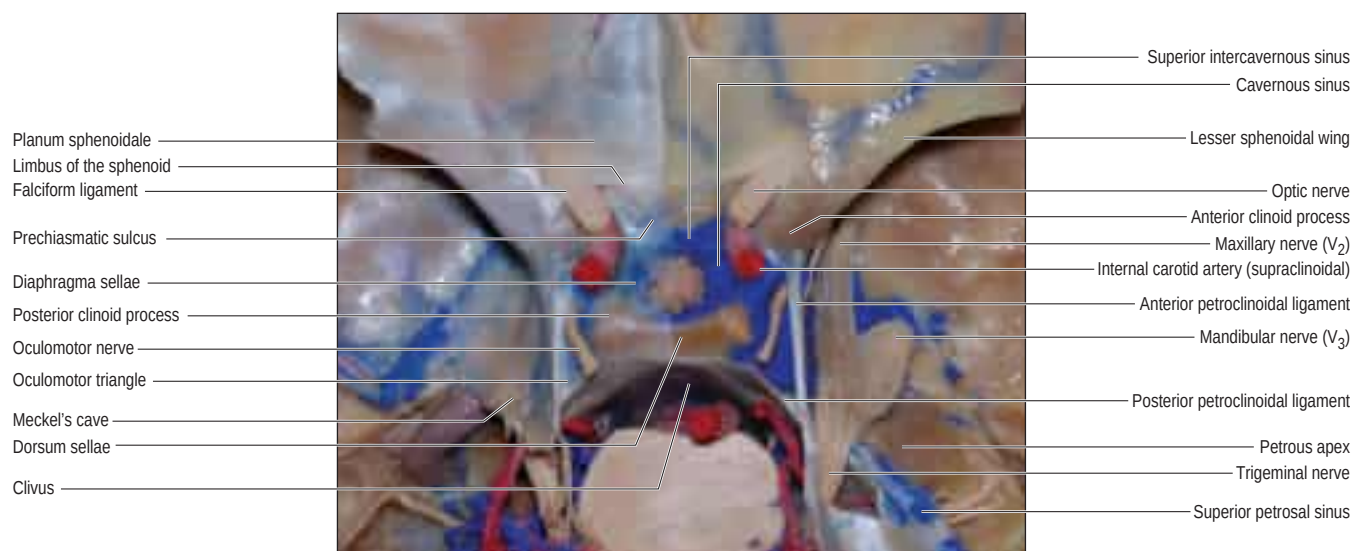


Fig. 28.9 The sellar region and adjacent middle and anterior cranial fossae, viewed from above, in a silicon-injected anatomical specimen. The dura has been removed on the right half of the specimen to show the roof of the cavernous sinus (oculomotor triangle), and the maxillary and mandibular nerves running on the floor of the middle fossa.

bone and turns forwards to the superior jugular bulb, lying posterior in the jugular foramen. Anteriorly, a thin plate of bone separates its upper part from the mastoid antrum and air cells. It connects with pericranial veins via mastoid and condylar emissary veins. Neurosurgical approaches to the lateral aspect of the posterior fossa (cerebellopontine angle) are classified as retrosigmoid, when the craniectomy is located just behind the sigmoid sinus, and presigmoid, when the mastoid bone in front of the sinus is drilled away to provide a more anterior corridor into the posterior fossa.

Superior petrosal sinus

These small, narrow sinuses drain the cavernous sinus into the transverse sinus on either side (see Figs 28.3, 28.5; Figs 28.9–28.10). Each leaves the posterosuperior part of the cavernous sinus and runs posterolaterally in the attached margin of the tentorium cerebelli, crosses above the trigeminal nerve to lie in a groove on the superior border of the petrous part of the temporal bone and ends by joining the transverse sinus at the point where this curves down to become the sigmoid sinus. The superior petrosal sinuses may receive cerebellar and brainstem veins, such as the superior petrosal vein, and rarely the inferior cerebral and tympanic veins; they connect with the inferior petrosal sinuses and the basilar plexus.

Inferior petrosal sinus

The inferior petrosal sinuses drain the cavernous sinus into the internal jugular veins (see Figs 28.5, 19.11). They begin at the posteroinferior aspect of the cavernous sinus and run back in the petroclival fissure, a groove between the petrous temporal and basilar occipital bones. On each side, the inferior petrosal sinus next passes through the antero-medial part of the jugular foramen, accompanied by a meningeal branch of the ascending pharyngeal artery, and descends obliquely backwards to drain into the superior jugular bulb. It sometimes drains via a vein in the hypoglossal canal to the suboccipital vertebral plexus.

The inferior petrosal sinuses are often plexiform. Each receives labyrinthine veins via the cochlear canaliculus and the vestibular aqueduct, and tributaries from the medulla oblongata, pons and inferior cerebellar surface. The venous spaces in the sphenopetroclival area, which are filled anteriorly by blood from the cavernous sinus, medially by blood from the basilar plexus, and laterally by blood from the superior petrosal sinus, typically drain into the inferior petrosal sinuses (Iaconetta et al 2003).

Cavernous sinus

The cavernous sinuses are two large venous plexuses that lie on either side of the body of the sphenoid bone, extending from the superior orbital fissure to the apex of the petrous temporal bone, with an average length of 2 cm and width of 1 cm (see Fig. 28.5). The cavity of the cavernous sinus is formed when the two layers of dura that cover the anterior aspect of the pituitary gland separate from each other at the lateral margin of the sella; the outer (endosteal) layer continues laterally to form the anterior (or sphenoidal) wall of the cavernous sinus, while the inner (meningeal) layer remains attached to the pituitary gland and runs back towards the dorsum sellae to form the medial

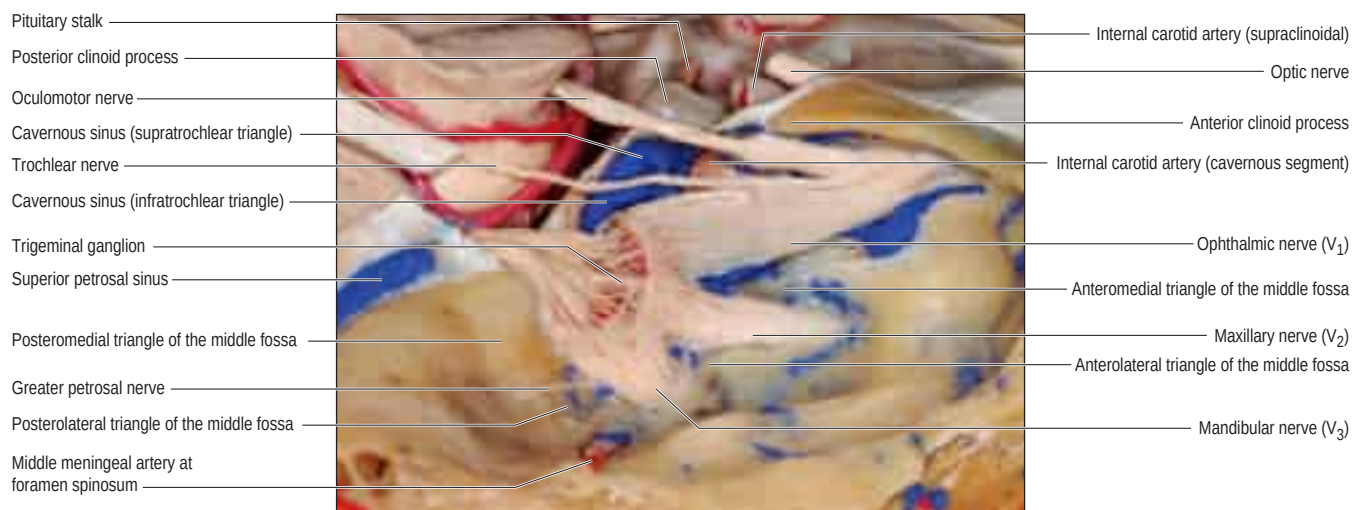
(or sellar) wall of the cavernous sinus. The posterior dural wall is located behind the dorsum sellae and upper clivus, and is shared with the basilar plexus. The lateral wall extends from the superior orbital fissure to the petrous apex, and separates the cavernous sinus from Meckel's cave, which is located inferior and posterior to the sinus. The roof of the cavernous sinus is formed by the dura lining the lower margin of the anterior clinoid process anteriorly (clinoid triangle), and the patch of dura between the anterior and posterior clinoids and the petrous apex, the oculomotor triangle (see Figs 28.9–28.10).

Each sinus contains the cavernous segment of the internal carotid artery, associated with a perivascular sympathetic plexus. The cavernous carotid artery has several portions, from proximal to distal: short ascending, posterior genu, horizontal, anterior genu. The anterior genu of the cavernous carotid artery continues as the paraclinoid segment of the carotid artery, which typically protrudes into the sphenoidal sinus cavity. The cavernous carotid gives off two major arterial branches: the meningohypophyseal and inferolateral trunks (see Fig. 28.10). The meningohypophyseal trunk arises typically from the posterior genu and gives off the inferior hypophyseal, tentorial and dorsal meningeal arteries. The inferolateral trunk arises a few millimetres distal to the meningohypophyseal and distributes around the nerves on the lateral wall of the sinus. (For more detailed descriptions of these branches, including their main variations, see Krisht et al (1994), Reisch et al (1996), Tekdemir et al (1998b).)

Several cranial nerves run forwards through the cavernous sinus to enter the orbit via the superior orbital fissure (see Fig. 28.10). The oculomotor and trochlear nerves and the ophthalmic division of the trigeminal nerve all lie in the lateral wall of the sinus. The abducens nerve enters the cavernous sinus by passing below the petrosphenoidal ligament within a dural tunnel (Dorello's canal) just behind the short ascending portion of the cavernous carotid artery, and then runs on the inferolateral side of the horizontal portion of the cavernous carotid, just medial to the ophthalmic nerve. The space between the trochlear and ophthalmic nerves at the lateral wall of the sinus, known as the infratrochlear or Parkinson's triangle, gives access to the horizontal cavernous carotid and posterior genu, meningohypophyseal trunk and abducens nerve. Unlike the ophthalmic division of the trigeminal nerve, the maxillary division of the trigeminal nerve does not run through the cavernous sinus or its lateral wall, but courses beneath the dura of the middle cranial fossa below the level of the cavernous sinus (see Fig. 28.10).

Several major venous spaces can be identified within the sinus in relation to the cavernous carotid artery: superior, posterior, inferior and lateral (modified from Harris and Rhoton 1976). The superior and inferior ophthalmic veins drain into the inferior space; the basilar plexus and inferior petrosal sinus drain into the posterior space; the superior petrosal sinus opens into the superior space; and the superficial middle cerebral vein, inferior cerebral veins and sphenoparietal sinus may drain into the lateral compartment. Veins traversing the emissary sphenoidal foramen, foramen ovale and foramen lacerum may also drain into the cavernous sinus. Less frequently, the central retinal vein and frontal tributary of the middle meningeal vein also drain into it.

A



B

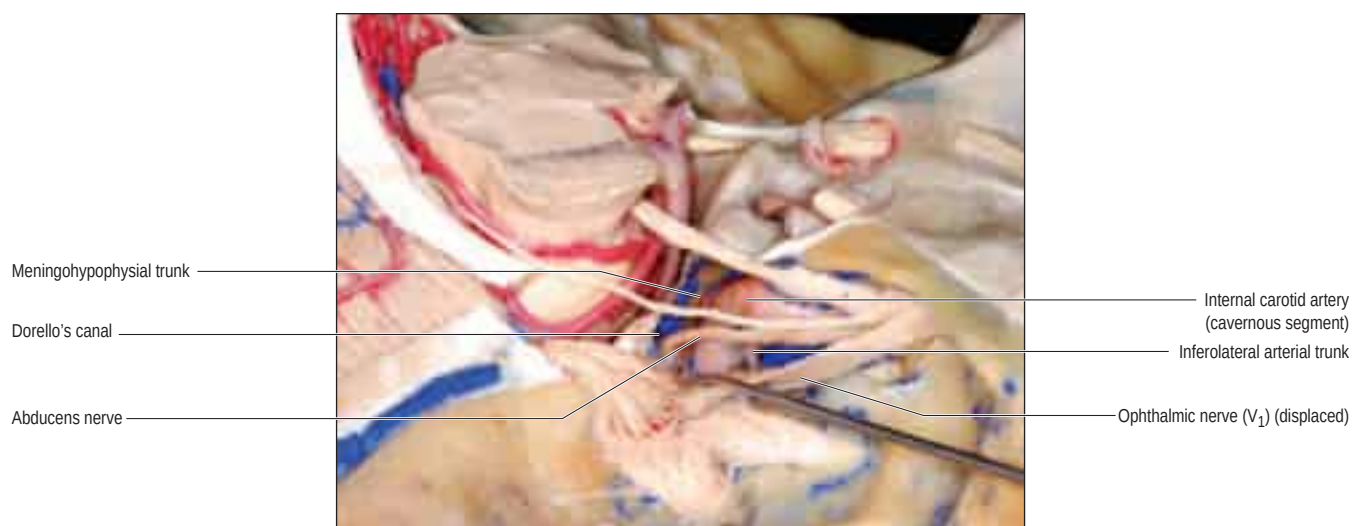


Fig. 28.10 **A**, A lateral view of the right middle cranial fossa and lateral wall of the cavernous sinus in a silicon-injected anatomical specimen. **B**, The ophthalmic nerve has been retracted inferiorly to show the inside of the cavernous sinus (abducens nerve, internal carotid artery and its branches) through the infratrochlear triangle.

Tumours can arise within the cavernous sinus (meningiomas, haemangiomas, schwannomas) or extend into the cavernous sinus from adjacent regions (typically, pituitary adenomas that invade the medial wall of the sinus). Transcranial microsurgical approaches enter the cavernous sinus through the lateral wall (infratrochlear triangle) or roof (oculomotor triangle) of the sinus. Recently introduced endoscopic endonasal approaches facilitate access to the cavernous sinus from its anterior (sphenoidal) and medial (sellar) walls. (For more detailed descriptions of the microsurgical anatomy of the cavernous sinus, see Dolenc (1987), Umansky et al (1994), Seoane et al (1998), Yasuda et al (2004), Dolenc and Rogers (2009).)

Carotocavernous sinus fistula and cavernous sinus thrombosis

Direct communication between the intracavernous portion of the internal carotid artery and the cavernous sinus, creating a carotocavernous sinus fistula (CCF), may occur as a result of either severe head trauma or aneurysmal vessel disease. The classic signs are ptosis, proptosis (which may be pulsatile), chemosis, periorbital oedema, and extraocular dysmotility causing diplopia secondary to a combination of third, fourth and sixth cranial nerve palsies. Hypo- or hyperaesthesia of the ophthalmic divisions of the fifth cranial nerve and a decreased corneal reflex may also be detected. There may be evidence of dilated, tortuous retinal veins and papilloedema. These changes can cause permanent blindness. CCFs are most commonly treated by passing a catheter up the carotid into the fistula, and then occluding it with dilatable balloons or flexible metal coils. Any spreading infection involving the upper nasal cavities, paranasal sinuses, cheek (especially near the medial canthus), upper lip, anterior nares, or even an upper incisor or canine tooth, may very rarely lead to septic thrombosis of the cavernous sinuses; infected thrombi pass from the facial vein or pterygoid venous

complex into the sinus via either ophthalmic veins or emissary veins that enter the cranial cavity through the foramen ovale. This is a critical medical emergency with a high risk of disseminated cerebritis and cerebral venous thrombosis.

Intercavernous sinuses

The two cavernous sinuses are connected by superior, inferior and posterior intercavernous sinuses, and the basilar plexus (see Figs 28.5, 28.9). The superior and posterior intercavernous sinuses lie in the anterior and posterior attached borders of the diaphragma sellae and they thus form a complete circular venous sinus. The inferior intercavernous sinuses are irregular and plexiform in nature, and commonly encountered in a surgical transnasal approach to the pituitary gland. Obliteration of any intercavernous sinus has no clinical consequences because all connections are valveless and the direction of flow in them is reversible.

Basilar venous plexus

The basilar sinus and plexus consist of interconnecting channels between the two layers of dura mater on the clivus (see Figs 28.5, 28.9). The basilar venous plexus interconnects the cavernous sinuses and inferior and superior petrosal sinuses, and joins the internal vertebral venous plexus.

Sphenoparietal sinus

The sphenoparietal sinuses are inconstant dural venous channels lying under the lesser wings of the sphenoid bone, near their posterior edge (see Fig. 28.5). Each sinus curves medially to open into the anterior part of the cavernous sinus, and receives small veins from the adjacent dura mater and sometimes the frontal ramus of the middle meningeal vein. Each may also receive connecting rami, in its middle course, from the superficial middle cerebral vein, temporal lobe veins and the

anterior temporal diploic vein. When these connections are well developed, the sphenoparietal sinus is a large channel. It has been suggested that the term sphenoparietal sinus should be abandoned on the basis that it is not an anatomical entity, but an artificial combination of two independent meningeal vessels, namely: the parietal portion of the anterior branch of the middle meningeal veins and the sinus of the lesser wing of the sphenoid (Ruíf *et al* 2004).

Occipital sinus

The occipital sinus is the smallest of the sinuses and it is typically larger in children. It lies in the attached margin of the falx cerebelli (see Fig. 28.5) and is occasionally paired. It commences near the foramen magnum in several small channels, one joining the end of the sigmoid sinus, and connects with the internal vertebral plexuses. It ends in the confluence of the sinuses (torcula).

Marginal sinus

The marginal sinus encircles the foramen magnum. It communicates anteriorly with the basilar plexus and with the occipital sinus posteriorly. It typically drains to the sigmoid sinus or jugular bulb by small sinuses and may connect extracranially to the internal vertebral venous plexus or the paravertebral or deep cervical veins in the suboccipital region.

Emissary veins

Emissary veins traverse cranial apertures and make connections between intracranial venous sinuses and extracranial veins. Some are relatively constant, while others may be absent. These connections are of clinical significance in determining the spread of infection from extracranial foci to venous sinuses, e.g. the spread of infection from the mastoid to the venous sinuses or from the paranasal sinuses to the cavernous sinus. They are also important because they may provide alternative drainage pathways in cases of venous sinus thrombosis.

The following emissary veins have been recognized. A mastoid emissary vein in the mastoid foramen connects the sigmoid sinus with the posterior auricular or occipital veins. A parietal emissary vein in the parietal foramen connects the superior sagittal sinus with the veins of the scalp. The venous plexus of the hypoglossal canal, which is occasionally a single vein, connects the sigmoid sinus and the internal jugular vein. A supracondylar emissary vein connects the sigmoid sinus and veins in the suboccipital triangle via the posterior condylar canal. A plexus of emissary veins (venous plexus of foramen ovale) connects the cavernous sinus to the pterygoid plexus via the foramen ovale. Two or three small veins traverse the foramen lacerum and connect the cavernous sinus and the pharyngeal veins and pterygoid plexus. A vein in the emissary sphenoidal foramen (of Vesalius) connects the cavernous sinus with the pharyngeal veins and pterygoid plexus. The internal carotid venous plexus, which passes through the carotid canal, connects the cavernous sinus and the internal jugular vein. The petrosquamous sinus is an emissary vein that courses over the lateral superior surface of the petrous part of the temporal bone. It arises from the dorsolateral portion of the transverse sinus, before its confluence with the superior petrosal sinus, and drains anteroinferiorly into the retromandibular vein and anteromedially into the pterygoid venous plexus. It usually disappears during the development of adult venous patterns in the last 3 months of prenatal life. A vein may traverse the foramen caecum and connect nasal veins with the superior sagittal sinus. An occipital emissary vein usually connects the confluence of sinuses with the occipital vein through the occipital protuberance, and also receives the occipital diploic vein. The occipital sinus connects with variably developed veins around the foramen magnum (so-called marginal sinuses) and thus with the vertebral venous plexuses; this pathway provides an alternative venous drainage when the jugular vein is blocked or tied. The ophthalmic veins are potentially emissary because they connect intracranial to extracranial veins.

Meningeal arteries and veins and diploic veins

Despite their names, the cranial meningeal arteries are predominantly periosteal (Fig. 28.11A). Their main targets are bone and haemopoietic marrow, and only some arterial branches are distributed to the cranial dura mater *per se*.

The branches of the meningeal vessels lie mainly in the endosteal layer of dura. In the anterior cranial fossa, the dura is supplied by the anterior meningeal branches of the anterior and posterior ethmoidal and internal carotid arteries and a branch of the middle meningeal artery. In the middle cranial fossa, it is supplied by the middle and accessory meningeal branches of the maxillary artery, a branch of the

ascending pharyngeal artery (entering via the foramen lacerum), branches of the internal carotid and a recurrent branch of the lacrimal artery. In the posterior fossa, the dura is supplied by the meningeal branches of the occipital artery (one enters the skull by the jugular foramen and another by the mastoid foramen), the posterior meningeal branches of the vertebral artery, occasional small branches of the ascending pharyngeal artery, which enter by the jugular foramen and hypoglossal canal, and the dorsal meningeal and tentorial arteries arising from the meningohypophyseal trunk. The anatomy of the meningeal arteries explains the vascular supply for tumours arising from the meninges (meningiomas) and the architecture of dural arteriovenous fistulae and malformations. (For detailed information on the surgical anatomy of the meningeal arteries, see Lasjaunias *et al* (2001), Martins *et al* (2005).)

Middle meningeal artery

The middle meningeal artery is the largest of the meningeal arteries. It arises from the first part of the maxillary artery in the infratemporal fossa and passes between the roots of the auriculotemporal nerve. It lies lateral to tensor veli palatini, then enters the cranial cavity through the foramen spinosum and runs in an anterolateral groove on the squamous part of the temporal bone, dividing into frontal and parietal branches (see Figs 28.10–28.11). The larger frontal (anterior) branch crosses the greater wing of the sphenoid and enters a groove or canal in the sphenoidal angle of the parietal bone (the sphenoparietal canal). It divides into branches between the dura mater and cranium; some branches ascend to the vertex. The parietal (posterior) branch curves back on the squamous temporal bone, reaches the lower border of the parietal bone anterior to its mastoid angle and divides to supply the posterior parts of the dura mater and cranium. These frontal and parietal branches anastomose with their fellows and with the anterior and posterior meningeal arteries.

Ganglionic branches supply the trigeminal ganglion and associated roots. The petrosal branch enters the hiatus for the greater petrosal nerve, supplies the facial nerve, geniculate ganglion and tympanic cavity, and anastomoses with the stylomastoid artery (El Khouly *et al* 2008). The superior tympanic artery runs in the canal for tensor tympani and supplies the muscle and the mucosa that lines the canal. Temporal branches traverse minute foramina in the greater wing of the sphenoid and anastomose with deep temporal arteries that supply temporalis. An anastomotic branch enters the orbit laterally in the superior orbital fissure, and anastomoses with a recurrent branch of the lacrimal artery; enlargement of this anastomosis is believed to account for the occasional origin of the lacrimal artery from the middle meningeal artery.

Accessory meningeal artery

The accessory meningeal artery may arise from the maxillary or the middle meningeal artery. It enters the cranial cavity through the foramen ovale, and supplies the trigeminal ganglion, dura mater and bone. Its main distribution is extracranial, principally to medial pterygoid, lateral pterygoid (upper head), tensor veli palatini, the greater wing and pterygoid processes of the sphenoid bone, the mandibular nerve and otic ganglion. It is sometimes replaced by separate small arteries.

Meningeal veins

Meningeal veins begin from plexiform vessels in the dura mater and drain into efferent vessels in the outer dural layer that connect with lacunae associated with some of the cranial sinuses. They include the middle meningeal and the diploic veins. Intracranial veins also communicate with extracranial vessels via emissary veins.

Middle meningeal vein (sinus)

The frontal (anterior) branch of the middle meningeal vein crosses the floor of the middle cranial fossa, from either foramen ovale or foramen spinosum, to the pterion, usually in the form of two parallel channels that accompany the middle meningeal artery. The vein subsequently passes cranially along the anterior margin of the parietal squama to empty into the venous lakes of the superior sagittal sinus. As they course under the most lateral aspect of the lesser sphenoidal wing, the anterior branches of the middle meningeal vessels are contained for a short distance within a bony canal, the sphenoparietal canal (of Trolard), which they leave to enter a groove on the internal surface of the parietal squama. The veins lie closer to the bone than the artery, and sometimes occupy separate grooves; they are particularly liable to tear in cranial fractures. Before entering the sphenoparietal canal, the anterior branch of the middle meningeal vein usually connects with the sinus of the lesser sphenoidal wing. The latter is connected medially with the anterior and superior aspect of the cavernous sinus by a channel that crosses over the superior ophthalmic vein to reach the cavernous sinus.

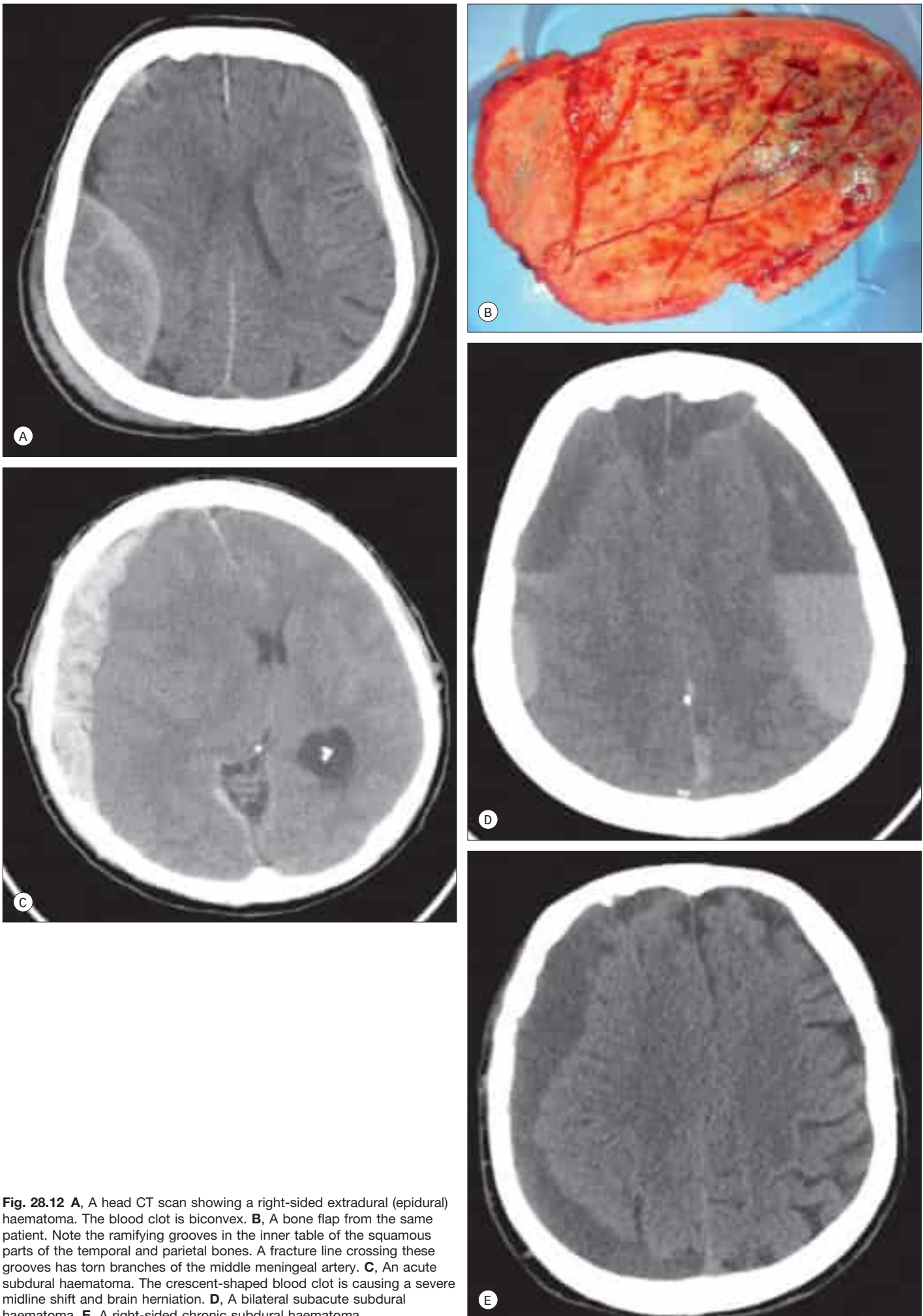
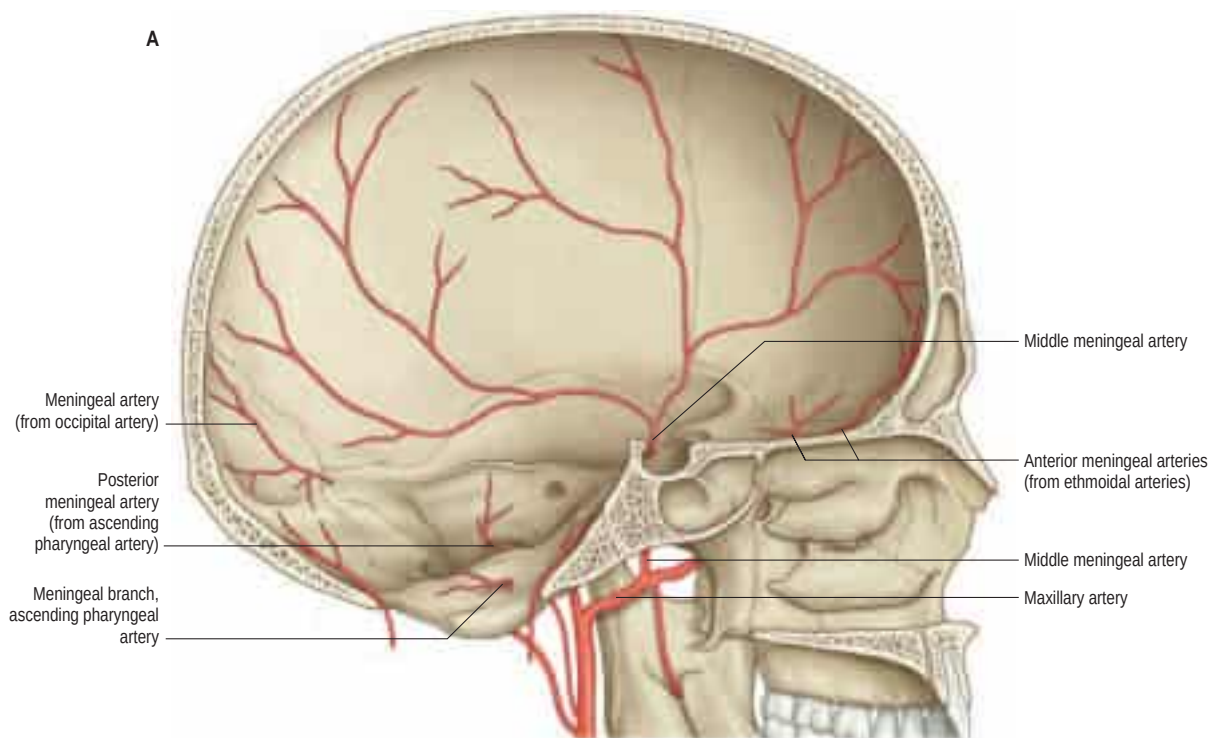


Fig. 28.12 **A**, A head CT scan showing a right-sided extradural (epidural) haematoma. The blood clot is biconvex. **B**, A bone flap from the same patient. Note the ramifying grooves in the inner table of the squamous parts of the temporal and parietal bones. A fracture line crossing these grooves has torn branches of the middle meningeal artery. **C**, An acute subdural haematoma. The crescent-shaped blood clot is causing a severe midline shift and brain herniation. **D**, A bilateral subacute subdural haematoma. **E**, A right-sided chronic subdural haematoma.

A



B

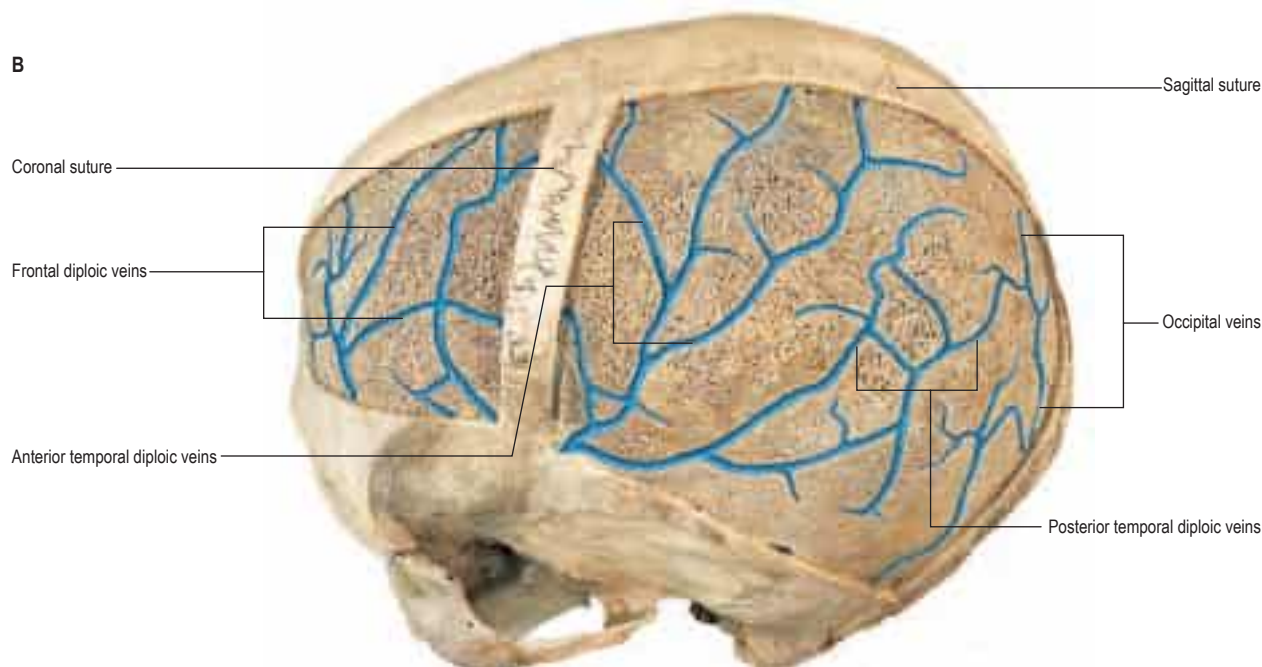


Fig. 28.11 **A**, The meningeal arteries. **B**, The diploic canals and veins after removal of the external table of the calvaria. (A, Adapted from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010. B, With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer, 2013.)

The parietal (posterior) trunk of the middle meningeal vein may traverse the foramen spinosum to end in the pterygoid venous plexus. The frontal trunk may also reach this plexus via the foramen ovale, or it may end in the sphenoparietal or cavernous sinus (see Fig. 28.5). The middle meningeal vein receives meningeal tributaries and small inferior cerebral veins, and connects with the diploic and superficial middle cerebral veins. It frequently bears arachnoid granulations.

Diploic veins

The diploic veins are large, thin-walled vessels that occupy channels in the diploë of the cranial bones (see Fig. 28.11B). Four main trunks are usually described; these are the frontal, anterior and posterior temporal, and occipital diploic veins. The frontal opens into the supraorbital vein and the superior sagittal sinus; the anterior temporal is mainly confined to the frontal bone, and opens into the sphenoparietal sinus, and one of the deep temporal veins through an aperture in the great wing of the sphenoid; the posterior temporal is situated in the parietal bone, and empties into the transverse sinus either through an opening at the

mastoid angle of the parietal bone or through the mastoid foramen; and the occipital, the largest of the four, is confined to the occipital bone, and may open externally into the occipital vein, or internally into either the transverse sinus or the confluence of the sinuses.

Sinus pericranii

Sinus pericranii is a rare condition involving congenital or acquired anomalous connections between an extracranial blood-filled nodule and an intracranial dural venous sinus via dilated diploic and/or emissary veins of the skull (Sheu et al 2002).

ARACHNOID AND PIA MATER

The arachnoid mater and the pia mater together are sometimes referred to as the leptomeninges. They are separated by the subarachnoid space and joined by trabeculae. They are composed of cells that share a common embryological origin from the mesenchyme that surrounds the developing nervous system. The outer layer of the arachnoid, the

dura–arachnoid interface, is formed from five or six layers of cells joined by numerous desmosomes and tight junctions. This layer forms a barrier that normally prevents permeation of CSF through the arachnoid into the subdural space. The central portion of the arachnoid is closely apposed to the outer layer, and is formed from tightly packed polygonal cells joined by desmosomes and gap junctions. The cells are more loosely packed in the inner layer of the arachnoid, where they intermingle with bundles of collagen continuous with the trabeculae that cross the subarachnoid space. The leptomeningeal cells are not surrounded by basement membrane, except where they are in contact with collagen in the inner layers of the arachnoid and on the deep aspects of the pia mater.

The anatomical relationships of the arachnoid and pia differ to some extent in the cerebral and spinal regions. The cerebral part of the arachnoid mater invests the brain loosely. It does not enter the sulci or fissures, other than the interhemispheric and Sylvian fissures, and it coats the superior surface of the pituitary fossa. The arachnoid is easily separated from the dura over the surface of the brain, but it is adherent to the adventitia of the internal carotid and the vertebral arteries where they enter the subarachnoid space. The arachnoid is reflected on to the surface of blood vessels and cranial nerves in the subarachnoid space, and this is the basis for microsurgical subarachnoid dissection to expose brain vessels and cranial nerves. The subarachnoid cisterns are cavities formed by the wide separation of the pia and arachnoid, and this is particularly common on the basal aspect of the brain, where the basal cisterns are located. Knowledge of the neural and vascular contents of each of the basal cisterns is of particular value to the neurosurgeon in the planning and execution of intracranial procedures. This is, however, beyond the scope of this chapter and the interested reader is directed in the first instance to [Yasargil \(1984\)](#).

Subdural haematoma

Separation of the arachnoid and dura mater requires little physical force, which means that damage to small bridging veins in the space can result in subdural haematoma after even relatively mild head trauma. In cases of chronic or subacute subdural haematomas, the accumulation is of relatively low pressure and seldom presents as a medical emergency; even sizeable accumulations may be tolerated on

a chronic basis with mild or no symptoms, and can be surgically drained through small openings in the skull (burr-holes). In many cases, there is some predisposing factor, such as cerebral atrophy or increased size of the underlying subarachnoid space. In cases of acute subdural haematoma, there is rapid accumulation of blood in the subdural space, typically after severe head trauma, which requires emergency neurosurgical drainage through a large craniotomy or craniectomy. The distinction between subdural and extradural haematoma on a CT scan relies on the anatomical features of the clot. Extradural collections tend to be lentiform in shape, reflecting the pressure required to separate the dura and periosteum. They will not pass deep to any major dural sinus and cannot extend along the falx cerebri or tentorium cerebelli. In contrast, acute subdural haematomas tend to be biconcave in shape and often follow the line of the dura along the falx or tentorium ([Fig. 28.12](#)).



Bonus e-book images and videos

Fig. 28.5 B–C, Digitally subtracted venograms, venous phase.

Fig. 28.12 A, A head CT scan showing a right-sided extradural (epidural) haematoma. **B**, A bone flap from the same patient. **C**, An acute subdural haematoma. **D**, A bilateral subacute subdural haematoma. **E**, A right-sided chronic subdural haematoma.

Video 28.1 3D surface rotation of the sella turcica in the horizontal plane.

Video 28.2 3D surface rotation of the sella turcica in the multiaxial plane.

Video 28.3 3D surface rotation of the sella turcica in the vertical plane.

KEY REFERENCES

- Browder J, Kaplan HA 1976 Cerebral Dural Sinuses and their Tributaries. Springfield, IL: Thomas.
A description of variations in the form of the superior sagittal and other venous sinuses.
- Dolenc VV 1987 The Cavernous Sinus. Vienna: Springer.
Professor Dolenc's description of his pioneering studies on the surgical anatomy of the cavernous sinus, which enabled the development of surgical approaches to this region.
- Fernandez-Miranda JC, Tormenti M, Latorre F et al 2012 Endoscopic endonasal middle clinoidectomy: anatomical, radiological, and technical note. *Neurosurgery* 71:ONS233–9; discussion ONS239.
A brief anatomical note that illustrates the surgical anatomy and variations of the middle clinoid.
- Iaconetta G, Fusco M, Samii M 2003 The sphenopetroclival venous gulf: a microanatomical study. *J Neurosurg* 99:366–75.
A detailed anatomical study of the abducens nerve, Dorello's canal and the so-called sphenopetroclival venous gulf.
- Lasjaunias P, Berenstein A, ter Brugge KG 2001 Surgical Neuroangiography. Berlin: Springer.
A comprehensive angiographic study describing the anatomical variations of the meningeal arteries.
- Martins C, Yasuda A, Campero A et al 2005 Microsurgical anatomy of the dural arteries. *Neurosurgery* 56:211–51.
A detailed anatomical study of the variations of the meningeal arteries and their surgical implications.
- Rhoton AL 2007 Cranial Anatomy and Surgical Approaches. Baltimore: Lippincott Williams & Wilkins.
An essential masterpiece in microsurgical neuroanatomy and surgical approaches developed by Professor Rhoton after 40 years devoted to the field.
- Yasargil MG 1984 Microneurosurgery, vol 1. New York: Thieme.
A monumental work in which Professor Yasargil describes the basal cisterns and their implications for microneurosurgery.

REFERENCES

- Ambekar S, Sonig A, Nanda A 2012 Dorello's canal and Gruber's ligament: historical perspective. *J Neurol Surg B Skull Base* 73:430–3.
- Barges-Coll J, Fernandez-Miranda JC, Prevedello DM et al 2010 Avoiding injury to the abducens nerve during expanded endonasal endoscopic surgery: anatomic and clinical case studies. *Neurosurgery* 67:144–54.
- Bauer DE, Youkilis A, Schenck C et al 2005 The falcine trigeminocardiac reflex: case report and review of the literature. *Surg Neurol* 63:143–8.
- Browder J, Kaplan HA 1976 *Cerebral Dural Sinuses and their Tributaries*. Springfield, IL: Thomas.
A description of variations in the form of the superior sagittal and other venous sinuses.
- Davidson JR, Mack J, Gutnikova A et al 2012 Developmental changes in human dural innervation. *Childs Nerv Syst* 28:665–71.
- Dolenc VV 1987 *The Cavernous Sinus*. Vienna: Springer.
Professor Dolenc's description of his pioneering studies on the surgical anatomy of the cavernous sinus, which enabled the development of surgical approaches to this region.
- Dolenc VV, Rogers L 2009 *Cavernous Sinus: Developments and Future Perspectives*. Vienna: Springer.
- El Khouly H, Fernandez-Miranda JC, Rhoton AL Jr 2008 Blood supply of the facial nerve in the middle fossa: the petrosal artery. *Neurosurgery* 62:ONS297–303.
- Fernandez-Miranda JC, Gardner PA, Snyderman C et al 2014 Clival chordomas: a pathological, surgical and radiotherapeutic review. *Head Neck* 36:892–906.
- Fernandez-Miranda JC, Tormenti M, Latorre F et al 2012 Endoscopic endonasal middle clinoidectomy: anatomical, radiological, and technical note. *Neurosurgery* 71:ONS233–9; discussion ONS239.
A brief anatomical note that illustrates the surgical anatomy and variations of the middle clinoid.
- Harris FS, Rhoton AL 1976 Anatomy of the cavernous sinus. A microsurgical study. *J Neurosurg* 45:169–80.
- Iaconetta G, Fusco M, Samii M 2003 The sphenopetroclival venous gulf: a microanatomical study. *J Neurosurg* 99:366–75.
A detailed anatomical study of the abducens nerve, Dorello's canal and the so-called sphenopetroclival venous gulf.
- Kakizawa Y, Abe H, Fukushima Y et al 2007 The course of the lesser petrosal nerve on the middle cranial fossa. *Neurosurgery* 61:15–23.
- Kaplan HA, Browder J 1976 Neurosurgical consideration of some features of the cerebral dural sinuses and their tributaries. *Clin Neurosurg* 23:155–69.
- Kawase T, Toya S, Shiobara R et al 1985 Transpetrosal approach for aneurysms of the lower basilar artery. *J Neurosurg* 63:857–61.
- Kerr RG, Tobler WD, Leach JL et al 2012 Anatomic variation of the optic strut: classification schema, radiologic evaluation, and surgical relevance. *J Neurol Surg B Skull Base* 73:424–9.
- Kılıç T, Akakin A 2008 Anatomy of cerebral veins and sinuses. In: Caso V, Agnelli G, Paciaroni M (eds) *Handbook on Cerebral Venous Thrombosis*. Front Neurol Neurosci 23:4–15. Basel: Karger.
- Krisht A, Barnett DW, Barrow DL et al 1994 The blood supply of the intracavernous cranial nerves: an anatomic study. *Neurosurgery* 34:275–9.
- Krmpotić-Nemanić J, Vinter I, Kelovizć Z et al 2005 Postnatal changes of the clivus. *Ann Anat* 187:277–80.
- Lasjaunias P, Berenstein A, ter Brugge KG 2001 *Surgical Neuroangiography*. Berlin: Springer.
A comprehensive angiographic study describing the anatomical variations of the meningeal arteries.
- Martins C, Yasuda A, Campero A et al 2005 Microsurgical anatomy of the dural arteries. *Neurosurgery* 56:211–51.
A detailed anatomical study of the variations of the meningeal arteries and their surgical implications.
- Morera VA, Fernandez-Miranda JC, Prevedello DM et al 2010 'Far-medial' expanded endonasal approach to the inferior third of the clivus: the transcondylar and transjugular tubercle approaches. *Neurosurgery* 66:211–19.
- Ozveren ME, Uchida K, Aiso S et al 2002 Meningovenous structures of the petroclival region: clinical importance for surgery and intravascular surgery. *Neurosurgery* 50:829–36.
- Paluzzi A, Fernandez-Miranda JC, Gardner P 2012 Retracing the etymology of terms in neuroanatomy. *Clin Anat* 25:1005–14.
- Reisch R, Vutskits L, Patonay L et al 1996 The meningohypophyseal trunk and its blood supply to different intracranial structures. An anatomical study. *Minim Invasive Neurosurg* 39:78–81.
- Rhoton AL 2007 *Cranial Anatomy and Surgical Approaches*. Baltimore: Lippincott Williams & Wilkins.
An essential masterpiece in microsurgical neuroanatomy and surgical approaches developed by Professor Rhoton after 40 years devoted to the field.
- Ruíz DSM, Fasel JHD, Rüfenacht DA et al 2004 The sphenoparietal sinus of Breschet: does it exist? An anatomic study. *Am J Neuroradiol* 25:112–20.
- Seoane E, Rhoton AL Jr 1999 Suprameatal extension of the retrosigmoid approach: microsurgical anatomy. *Neurosurgery* 44:553–60.
- Seoane E, Rhoton AL Jr, de Oliveira E 1998 Microsurgical anatomy of the dural collar (carotid collar) and rings around the clinoid segment of the internal carotid artery. *Neurosurgery* 42:869–84.
- Sheu M, Fauteux G, Chang H et al 2002 Sinus pericranii: dermatologic considerations and literature review. *J Am Acad Dermatol* 46:934–41.
- Tauber M, van Loveren HR, Jallo G et al 1999 The enigmatic foramen lacerum. *Neurosurgery* 44:386–91.
- Tekdemir I, Tuccar E, Aslan A et al 1998a The jugular foramen. A comparative radioanatomic study. *Surg Neurol* 50:557–62.
- Tekdemir I, Tuccar E, Cubuk HE et al 1998b Branches of the intracavernous internal carotid artery and the blood supply of the intracavernous cranial nerves. *Ann Anat* 180:343–8.
- Umansky F, Valarezo A, Elidan J 1994 The superior wall of the cavernous sinus: a microanatomical study. *J Neurosurg* 81:914–20.
- Yasargil MG 1984 *Microneurosurgery*, vol 1. New York: Thieme.
A monumental work in which Professor Yasargil describes the basal cisterns and their implications for microneurosurgery.
- Yasuda A, Campero A, Martins C et al 2004 The medial wall of the cavernous sinus: microsurgical anatomy. *Neurosurgery* 55:179–89.